Video-Derived Side-Channel Attacks on ECG-Based Cryptography

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Abstract

Electrocardiogram (ECG) signals are increasingly proposed as a robust and secure basis for biometric cryptography and authentication, grounded on the assumption that these physiological signals are "unobservable" without direct physical contact. This paper aims to challenge this assumption by investigating a non-contact side-channel attack pipeline that utilizes consumer-grade video to compromise ECG-based cryptographic schemes. Our proposed framework integrates remote photoplethysmography (rPPG) to extract cardiac signals from facial video, a Generative Adversarial Network (CardioGAN) to synthesize corresponding ECG waveforms, and a range of recreated test cryptographic models for exploitation. We evaluate several rPPG architectures, including CHROM, PhysNet, and TS-CAN on existing datasets and smartphone footage. Our results indicate that while TS-CAN achieves a usable heart rate Mean Absolute Error (MAE) of 10.09 BPM on realistic footage, significant morphological challenges remain in the subsequent ECG synthesis step, particularly regarding T-wave and R-peak differentiation. Testing against existing schemes like Heart-to-Heart (H2H) and H2K reveals that while current reconstruction inaccuracies limit consistent real-time cracking, the underlying security premise of ECG unobservability is increasingly becoming more fragile. This work highlights a critical vulnerability in the biometric security landscape and advocates for the adoption of stronger defense mechanisms such as multi-factor authentication (MFA) and noise-resilient cryptographic designs.

I. Introduction

BIOMETRIC cryptography has emerged in research as an alternative to traditional password-based security mechanisms, offering both the potential for stronger identity verification and a vector for key generation. Among these biometrics, electrocardiogram (ECG) signals have gained increasing attention in recent research. ECG data is particularly appealing for authentication and cryptographic schemes, purportedly because it is continuous, difficult to forge directly, and inherently linked to an individual's physiological characteristics.

Current research suggests that ECG data can be utilized effectively in real time for cryptographic schemes and authentication. These systems typically assume that ECG signals can only be obtained through direct physical contact with a sensor, such as a wearable device or medical-grade electrode. However, recent developments in remote photoplethysmography (rPPG) suggest a potential side-channel, where subtle skin-color changes captures via consumer-grade video capture may leak sufficient cardiac information to reconstruct an individual's underlying ECG.

If ECG-correlated signals can be reconstructed or approximated from video data, then an attacker may gain the capability to infer ECG features without physical contact in real-time. This creates an attack vector where an adversary can collect video footage of a victim and use it to derive biometric information assumed to be private. As a result, cryptographic schemes or real-time authentication systems that rely on the secrecy or unobservability of ECG signals would be rendered fundamentally vulnerable.

The core issue would not merely be a weakness in implementation, but a conceptual flaw in the underlying ECG-based cryptographic systems. If indirect acquisition of ECG data is feasible, then these systems may be susceptible to replay attacks, impersonation, or cryptographic key reconstruction without the victim's knowledge. This problem has significant implications for both academic research and real-world deployments of biometric-based security systems.

The primary objective of this work is to investigate the practical feasibility of an end-to-end attack pipeline capable of cracking ECG-based cryptographic schemes in real time using only video-derived data. Rather than assuming privileged access to medical sensors or wearable devices, the proposed system operates under a realistic adversarial model in which only video data of the target is available. By doing so, we aim to demonstrate that ECG-based cryptographic security can be compromised without physical proximity or direct interaction.

Furthermore, this work contributes to the broader discussion of ECG-based cryptographic schemes by breaking several modern schemes reported in the recent literature. By exposing potential vulnerabilities, we aim to encourage the development of cryptographic schemes that do not rely solely on the assumed secrecy of physiological signals, as well as safer authentication schemes such as two-factor authentication (2FA).

II. Background

In this section, we will define two distinct methodologies for measuring cardiac signals that are integral to our project and their relevant attributes that are utilized as input for cryptographic schemes. We will also give an overview of methods for recording these signals.

A. Cardiovascular Biosignals

Cardiovascular biosignals can be defined as physiological measurements of time-based cardiac activity. The relevant biosignals for our study are the electrocardiogram (ECG internationally; EKG in the US) and photoplethysmogram (PPG). There are many other biosignals such as seismocardiogram (SCG) and ballistocardiogram (BCG) that we will not explore in this report.

B. Electrocardiogram

Electrocardiograms (ECG) are a measure of the electrical activity of the heart. The activity is a consequence of the depolarization (contraction) and repolarization (relaxation) of cardiac muscle cells during every heartbeat. This cyclic process results in the ECG signal and its distinct features. The ECG signal is composed of the P and T wave, the Q and S negative peaks, and the R positive peak (PQRST) [1].

Several studies have showcased that analysis of these features is unique among different individuals. For example, the time differences between consecutive heartbeats (R-R peaks), or Inter-Pulse-Intervals (IPIs), are considered unique enough between individuals to be used for biometric identification [1]. Similarly, there are many other relationships between the PQRST features that are considered unique to each individual [1]. At the same time, it is important to recognize that these signals can change across time, both in an immediate way (across minutes to hours) and across longer periods of years. As a result, ECG signals are not considered time-invariant.

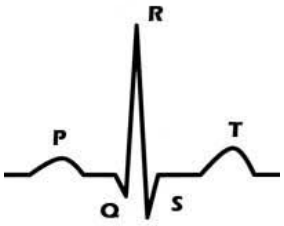


Fig. 1: Standard ECG waveform morphology [2]

Raw ECG signals captured directly from body surface electrodes are susceptible to numerous noise sources and artifacts that can degrade signal quality and obscure the PQRST waveform components. The primary sources of contamination include muscle artifacts from voluntary and involuntary muscle contractions and baseline wander caused by respiration [3]. Additional artifacts arise from electrode polarization, patient tremors, and external vibrations [3]. These interference sources can have amplitudes comparable to or exceeding the ECG signal itself, making the detection of critical features unreliable. Raw ECG signals, therefore, exhibit irregular baselines and obscured waveform morphology that compromise biometric feature extraction for cryptographic applications.

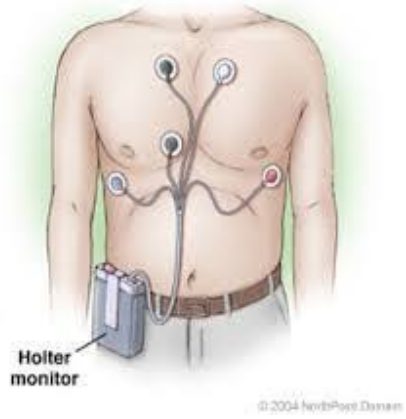
Cleaned ECG signals undergo preprocessing to eliminate these artifacts and excess noise. ECG signals commonly undergo high-pass/low-pass filtering and wavelet decomposition as preprocessing steps, but more advanced denoising algorithms are also used.

1) ECG Acquisition Methods: ECG is primarily measured by electrodes, a non-invasive sensor capable of outputting measurable electrical current from bio-electric activity happening inside the human body. In clinical settings, a 12-lead ECG is recorded by placing 10 electrodes on different parts of the body (e.g., 4 on the limbs, 6 on the chest). The 12 ”leads” are not physical electrodes but derived voltage differences calculated from those 10 electrodes. The 12-lead electrodes are manufactured to record the ECG signal with high-fidelity and provide quick and accurate snapshots able to detect arrhythmias, heart attacks or other cardiac issues. There are also simpler ECG recording methods involving 3 or 5 leads. One such is a Holter monitor, a portable device worn by a patient continuously for 24 to 48 hours [4].

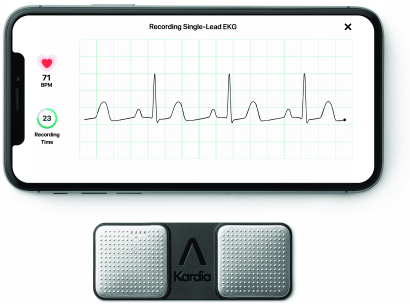
Innovative consumer devices have recently been introduced that use 1-lead sensors such as the KardiaMobile [5] and smartwatches. However, these require a completion of the circuit by touching a part of the device with the other hand.



(a) 12-lead ECG Machine [6]



(b) ECG Holter Monitor [7]



(c) Single-lead ECG device [8]

Fig. 2: Comparison of various ECG monitoring devices.

C. Photoplethysmogram (PPG)

Photoplethysmogram or PPG is a measure of the variation in blood flow in the vessels [9]. The fundamental principle behind the measurement of PPG is that blood absorbs light more effectively than its surrounding tissue. When specific parts of the body such as fingertips or the face are illuminated (usually by LED lights), more blood flow will result in more light absorption. A photodetector (photodiode) will then measure the intensity of light either reflected or transmitted through the tissue. When the heart is beating, more blood flows into the area which results in higher light absorption and less light measured by the detector. The photodetector will convert this into a PPG waveform that we can analyze. A PPG signal/waveform is composed of AC and DC components, systolic and diastolic peaks, and the dicrotic notch.

The alternating current (AC) component represents the consistent pulse portion of the PPG signal that varies with each cardiac cycle. This component is synchronized with the heartbeat and reflects the changes in blood volume caused by the rhythmic pumping action of the heart. The AC component contains valuable information about heart rate, heart rate variability, and vascular dynamics. It typically has a much smaller amplitude compared to the DC component and is the primary component analyzed for extracting cardiovascular parameters.

The direct current (DC) component represents the baseline portion of the PPG signal. This component reflects the average blood volume in the tissue, as well as light absorption by other tissues such as skin, bone, muscle, and venous blood. The DC component is relatively constant but can slowly vary due to factors such as respiration, vascular action, body temperature changes, and movement. While it changes more slowly than the AC component, it is important for normalizing the signal and can provide information about tissue perfusion and vascular tone.

The systolic peak is the maximum point of the PPG waveform, corresponding to the peak of ventricular contraction. The diastolic (or dicrotic) peak is a smaller secondary peak that occurs after the systolic peak, during the diastolic phase when the heart is relaxing and refilling. The time interval between systolic peaks is used to calculate heart rate. The dicrotic notch is a distinctive downward deflection in the PPG waveform that appears between the systolic peak and the diastolic peak.

Raw PPG signals acquired directly from sensors are inherently contaminated with various sources of noise and artifacts that can significantly compromise signal quality and subsequent analysis. The most prominent sources of this contamination include motion artifacts from sensor or body movement and ambient light contamination. These artifacts obscure the characteristic cardiac waveform morphology presented above, making reliable feature extraction and biometric authentication challenging. As a result, PPG signals undergo systematic preprocessing to remove noise while preserving physiological information, resulting in a more suitable waveform for cryptographic key generation and biometric matching [10].

PPG signal preprocessing typically employs bandpass filtering to retain cardiac frequency components while eliminating baseline drift and high-frequency noise. However, different artifacts can require varying levels of preprocessing steps, from Empirical Mode Decomposition (EMD), Savitzky-Golay filtering, Independent Component Analysis, and wavelet decomposition, which will not be discussed in this paper.

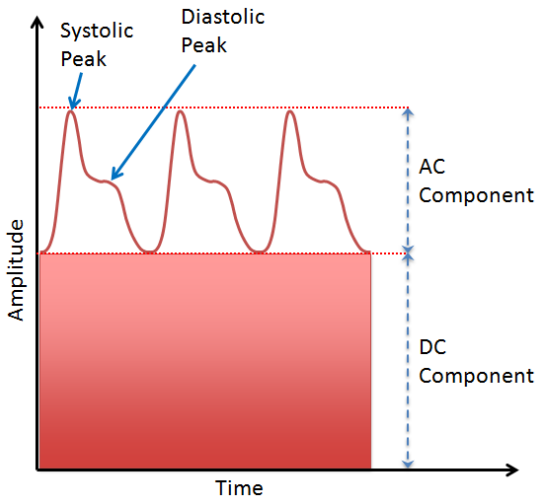
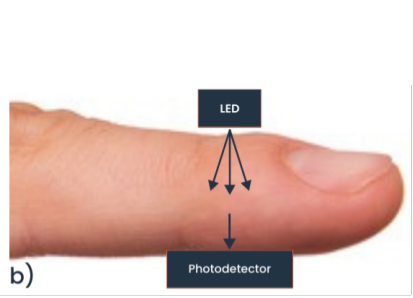


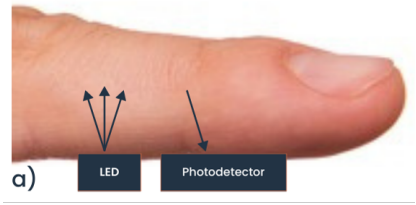
Fig. 3: Standard PPG waveform [11]

1) PPG Recording Methods:

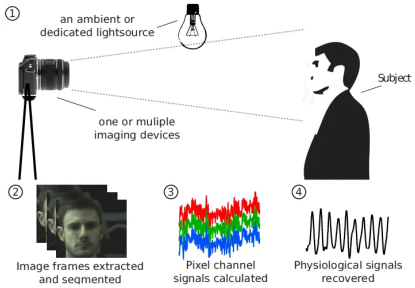
- Transmittance Mode: The light is transmitted through the tissue, which is configured between the light source and the detector. The detector measures how much light was not absorbed by the blood and bone. This method is used in hospitals through finger or earlobe clips.
- Reflectance Mode: This is the most common PPG method found in consumer electronics such as smartwatches or earbuds. The light source and the detector are placed on the same side of the skin and the detector measures the amount of light reflected by the blood vessels.
- Imaging Photoplethysmography (iPPG): This method is non-contact, and it involves using a camera in controlled environments to map cardiovascular signals by analyzing subtle changes in skin color. By detecting minute variations in light absorption across a video feed, iPPG allows for remote, non-invasive measurement of heart rate, breathing, and tissue blood perfusion without sensors [12].
- **Remote PPG (rPPG):** This is a more recent contacless technology that utilizes camera sensors to detect subtle color changes in the skin (typically the face). The light source in this case is the environmental light or specialized infrared illumination. Because rPPG relies on a camera working at a distance to the patient and can be affected by several factors such as light fluctuations, motions and hardware noise, is it the method with the lowest Signal-to-Noise Ratio (SNR). However, due to the desirability of no-contact in our threat model, we will utilize advances in rPPG within our attack pipeline. We will cover existing work on rPPG in the next section and rPPG attacks in Section III.



(a) Transmittance [13]



(b) Reflectance [13]



(c) rPPG [14]

Fig. 4: Comparison of various PPG modalities.

D. Wireless Body Area Networks

Wireless Body Area Networks (WBANs) represent an emerging paradigm in healthcare monitoring, integrating a network of wearable and implantable sensors distributed across the human body to collect continuous physiological data. These networks enable the simultaneous acquisition of various biosignals, including ECG, heart rate, temperature, blood oxygen saturation, and movement data, which can be wirelessly transmitted for real-time analysis. The concept of WBANs has gained significant traction in recent years due to advances in miniaturization, wireless communication protocols, and sensor technology [15]. These innovations have made it increasingly feasible to deploy multiple medical devices on or within the body. This convergence has

given rise to what is often termed integrated medical device ecosystems [16]. These comprehensive systems integrate multiple sensors that work cooperatively to provide holistic patient monitoring.

The miniaturization that enables WBAN deployment, however, introduces significant computational and resource constraints. Smaller sensor nodes inherently possess limited processing power and memory capacity. This miniaturization is beneficial in that it facilitates the integration of sensors into everyday objects. However, these same size reductions mean that traditional cryptographic algorithms and data processing methods become impractical or even impossible to implement [17]. The energy cost of complex encryption, frequent wireless transmission, and continuous signal processing forces trade-offs between security and device longevity. Consequently, researchers and engineers have focused on developing lightweight algorithms to more carefully balance functionality with limitations imposed by miniaturization.

WBANs often collect ECG and PPG signals through strategically placed sensors. Traditionally, these signals have been used solely for diagnostic purposes. However, as these medical devices become interconnected, challenges with managing and securing the data generated have become more pronounced. The integration of multiple sensors into a unified WBAN inherently introduces vulnerabilities and privacy risks [17]. In response to these data security challenges, research in the realm of biocryptography has reframed the problem into a solution by leveraging the very biosignals collected by WBANs [18]. Biocryptographic approaches utilize the inherent randomness, temporal variability, and individuality of biosignals such as ECG and PPG to generate encryption keys or authenticate devices. Because physiological signals exhibit unique characteristics and change dynamically over time, these signals provide a source of entropy and identity verification that is difficult to replicate or forge. This approach serves as an avenue to protect medical data transmitted across the resource-constrained body area network devices where conventional key generation algorithms are too computationally expensive or energy-intensive to implement [17].

E. rPPG Methods

1) Traditional Unsupervised: Unsupervised rPPG methods extract heart rate information from video data without requiring labeled physiological signals during training. These approaches are grounded in physical signal modeling based on the notion that cardiac-induced blood volume changes cause subtle variations in skin color that can be detected through video cameras. Unsupervised methods leverage optical models of light absorption combined with computer vision techniques, specifically facial recognition algorithms, to localize regions of interest for signal extraction. Common approaches in this category include signal decomposition methods, which separate cardiac signals from noise, and color channel processing techniques that use relationships between color channels to enhance the cardiac signal.

Color Chrome (CHROM) is a specific unsupervised model that was used in our research. CHROM is an unsupervised rPPG method that uses color channel processing. The CHROM rPPG pipeline is shown in figure [19]. CHROM works by first using computer vision to detect a region of interest, specifically a face, to create a mask that the color processing can be conducted on. The second step involves spatial signal acquisition, where the RGB color values are extracted and averaged across all pixels within the masked facial region for each video frame, resulting in three temporal signals corresponding to the red, green, and blue channels. The third step addresses motion artifact reduction through temporal normalization and bandpass filtering to remove low-frequency trends and high-frequency noise that do not correspond to physiological signals. The fourth step involves pulse signal construction, where CHROM applies a linear combination of the RGB signals while suppressing motion artifacts to construct the PPG signals and heart rate. The fifth and final step is pulse feature extraction, where frequency domain analysis is applied to the constructed pulse signal to identify the dominant frequency peak.

2) Supervised Deep Learning: Supervised deep learning rPPG methods use neural networks trained on labeled video data paired with ground truth physiological signals to learn robust pulse extraction. These methods employ end-to-end architectures that typically consist of input processing layers, feature extraction modules, and output layers that predict physiological signals. Supervised deep learning approaches have demonstrated superior performance compared to unsupervised methods in situations involving subject movement, challenging lighting conditions, and diverse skin tones, as they can learn to adapt to various sources of noise and variability present in real-world scenarios.

PhysNet is a supervised rPPG model used in our research. PhysNet is a prominent architecture that utilizes a 3D convolutional neural network (CNN) to process video segments and extract temporal features across multiple frames simultaneously [20]. PhysNet works by employing an encoder-decoder structure with 3D convolutions that capture both spatial information within frames and temporal dependencies across frames. This allows it to learn features that correlate with blood volume changes in a more robust way. The network is trained using a combination of loss functions that compare the predicted pulse signal to ground truth measurements, enabling it to learn complex mappings between facial video appearance patterns and underlying cardiac activity.

Temporal Shift Convolutional Attention Network (TS-CAN) is another supervised rPPG model used in our research. TS-CAN was developed after the creation of PhysNet and introduced several architectural improvements to enhance pulse signal extraction [21]. TS-CAN differs from the aforementioned models in that it uses temporal shift modules. These modules enable more efficient temporal modeling by shifting feature maps along the temporal scale. This approach allows the network to capture temporal dependencies across frames whilst simultaneously being potentially lighter in nature compared to 3D-CNN modules.

III. Literature Review

A. Evolution of Remote PPG Attacks

PPG-based biometric authentication has been considered secure due to PPG's unobservable nature and its unique user identifier. Several patents have even been filed by Samsung [22], Nymi [23], and Apple [24] that take advtanage of this blood flow measurement method.

Significant research on PPG attacks has largely required direct access to a victim's leaked PPG signals, either through malicious sensors or close physical contact, placing significant practical constraints on realistic attack scenarios. Presenting a threat model where the victim's PPG signal is assumed, severely limits the significance of the attacks. The advent of remote PPG (rPPG), however, has enabled substantial work on remote handling of PPG and attacks on existing biomeric systems.

Li et al. [25] present a novel spoofing attack against PPG-based biometric authentication systems, demonstrating that an adversary requires nothing more than a publicly available HD video of a victim's face to compromise such systems. This threat model is much stronger considering attackers can find videos of victims in several online social media platforms like TikTok, YouTube and Facebook with increasing ease. The authors propose SigR, a GAN-based signal restoration model that translates rPPG signals into PPG-like waveforms. The authors conducted varying experiments on the UBFC-PHYS dataset from several states (resting, talking, calculating) and differing video qualities such as frame rate, frame size, bit-rate and use of beauty filter. SigR achieves superior false acceptance rates (FAR) than previous models, proving that rPPG attacks pose a major threat to PPG-based authentication.

This work has since inspired further extensions of PPG and rPPG-based attacks. Yao et al. [26] present a novel backdoor attack that is aimed at PPG-based biometric authentication. Their system utilizes intentional waveform variations (backdoor) stealthily embedded in PPG signals used as input for PPG-based authentication systems. [27] introduces a comprehensive database of rPPG spoofing attacks consisting of 25 unique attack scenarios. The forged PPG signals are created through various methods including illumination, motion and pixel variations for deception in liveness detection systems.

Wang et al. [28] introduce FakePPG, a novel physical-domain sensor spoofing attack which utilizes a low-cost Liquid Crystal Modulator (LCM) to recreate the pulsating light-absorbing characteristics of human skin. This approach becomes extremely sophisticated when combined with rPPG, providing a method to re-materialize signals captured remotely from rPPG and present them to a hardware sensor as a physical pulse. In their evaluation, the authors show that their method deceives both wearable sensors and camera-based rPPG applications.

The field of rPPG-based attacks is still in its infancy with many more possibilities left to explore and systems to break. However, although our work depends on rPPG methods, we are primarily interested in delving into the uncharted territory of breaking ECG-based cryptographic systems in real-time.

B. ECG synthesis and spoofing

Extensive work has gone into ECG synthesis, primarily for improving medical diagnosing of heart failure. In this case, synthesis refers to the artificial generation or reconstruction of the biological signal in contrast to recording it directly from the person. Early approaches in [29] were centered on generating realistic ECG signals as the sum of Gaussian curves (bell-shaped curves) based on features manually extracted from the patient's ECG. This method allowed for high-quality and dynamic ECG synthesis. Current models are much more sophisticated and are based on deep generative models such as Variational Autoencoder [30] and diffusion models [31]. Furthermore, Generative Adversarial Networks (GANs) have risen as the dominant method for synthesizing ECG as part of an effort to improve the quality of existing data. CardioGAN is an example of a GAN-based model that we used in our attack pipeline [32].

Similar to PPG-based authentication attacks, ECG-based authentication attacks have emerged as a growing field of research in security. Karimian et al. demonstrated a novel spoofing attack against against the Nymi band, a commercial wristband that utilized ECG authentication [33]. In the paper, the authors developed a model that translates ECG recordings from a nearby portable ECG monitor to the structure required by the Nymi band. Karimian et al. expanded on this research by developing attacks that synthesize a victim's ECG signals. These techniques involved utilizing another subject's ECG recorded from the same device or by converting a partial ECG signal into the full reading required. They tested these spoofing attacks against several ECG authentication systems. Both of these papers show promising steps in the field of ECG spoofing but suffer several limitations. The threat model can be viewed as severely unrealistic, requiring either close proximity to the victim or a supervised training dataset including the victim's ECG traces. If attacks have access to such a dataset, then perhaps advanced spoofing methods aren't even required.

C. Cross-Modal Synthesis and Cryptographic Implications

A recent and highly relevant study, "Biosignal Authentication Considered Harmful Today" by Krish et al. [34] extended [25] by evaluating the vulnerability of ECG-based authentication systems. The authors challenged the assumption underlying ECG-based authentication systems that these physiological signals are inherently difficult for an attacker to obtain or reproduce. Their work demonstrates that an adversary who gains access to PPG can synthesize another signal type, such as an ECG, that is sufficiently realistic to bypass authentication systems.

Using this approach, the study demonstrated that synthesized ECG signals derived from PPG can successfully compromise a variety of state-of-the-art authentication systems. The authors even extended the attack by extracting rPPG signals directly from video recordings of a target individual utilizing the CHROM method and showed that even passive video observation could provide enough physiological information to generate spoofed ECG signals capable of bypassing authentication mechanisms. These findings highlight a fundamental limitation of biosignal-based authentication that cardiovascular signals cannot be assumed to remain private or difficult to replicate.

This work has been instrumental to the inspiration and ideation for this Capstone project. We will mainly be extending their work to attempt to break ECG-based cryptographic systems and investigate the feasiblity of performing the attack in real-time. We have worked heavily within their accompanying open-source repository titled "biosignal-auth-harmful". When attempting to replicate their results we encountered several portability issues running their code on our local machines as well as several code bugs. We made several improvements to the code through Git commits which were integrated after our pull request was approved. Their system and evaluation code is publicly available at the following repository: https://github.com/Ethos-lab/biosignal-auth-harmful.

D. ECG-based Cryptographic Schemes

Several previous works delve into breaking biological authentication schemes. However, no comprehensive work showcases ECG-based cryptographic cracking on a comprehensive basis. This section presents a variety of cryptographic key generation and encryption schemes that utilize rule-based and machine-learning-based methodologies that this paper attempts to crack.

1) IPI-Based Key Agreement:

Rostami et al. introduced Heart-to-Heart (H2H), an ECG-based cryptographic pairing protocol designed to establish secure communication between implantable medical devices (IMDs) and external programmers [35]. The protocol addresses a critical security vulnerability in contemporary IMDs, which often lack robust authentication mechanisms. Unlike previous ECG-based pairing schemes that suffered from serious cryptographic flaws [36], H2H provides the first physiologically-based IMD device pairing protocol. The system enforces a "touch-to-access" policy through the use of ECG as a means of preventing remote attacks while maintaining usability in emergency medical situations.

The H2H protocol operates by extracting randomness from inter-pulse intervals (IPIs). The system specifically utilizes the four least significant bits of quantized IPI values, which the researchers confirmed to be independently and identically distributed random variables through analysis of extensive patient databases. These randomized bits, denoted IPI4 values, are simultaneously sampled. To accommodate measurement noise inherent in physiological signals, H2H employs a novel application of the Neyman-Pearson Lemma for statistical hypothesis testing rather than relying on simple Hamming distance comparisons used in prior work. This approach accounts for the fact that different bit positions in the IPI4 values exhibit varying error rates and weights them accordingly during pairing. The authentication decision is made by computing an error value vector that captures the total number of errors in each bit position and comparing it against an optimally computed threshold.

The H2H system functions as a cryptographic pairing protocol that establishes a unified shared key following successful authentication between two ECG-reading devices. The protocol architecture consists of two phases. First, a secure-channel setup phase uses TLS to establish an encrypted but unauthenticated channel between devices. Second, an authentication phase employs a commitment/decommitment scheme. The protocol exploits the time-varying nature of ECG signals to provide forward security, meaning that old authentication values should not be able to reveal future ones. The authentication succeeds when the ECG readings are within the statistically determined threshold, allowing the devices to establish shared cryptographic keys for secure subsequent communication. The system is specifically designed to accommodate the tight computational and power constraints of IMDs while tolerating the inherent noise in physiological measurements.

Rostami et al. evaluated H2H through statistical analysis using patient data from the MIT-BIH Arrhythmia Database, the PTB Database, and the MGH/MF Waveform Database. The researchers confirmed that approximately four truly random and statistically independent bits could be extracted from each heartbeat, and that these bits passed all NIST statistical randomness tests. For their implementation, the team set a false-negative requirement of 10^4 and chose to collect n=15 IPI4 values, each requiring 11-15 seconds of ECG measurement. This configuration achieved a false positive rate of $2.7{\times}10^{-}9$, which exceeds the security level of a typical eight-digit RSA SecurID token.

Similarly, Zhang et al. proposed the H2K framework, a heartbeat-based key generation system that leverages IPIs from ECG and signals as a biometric-based alternative for securing wireless body area networks (WBANs) [37]. This approach again exploits the inherent randomness of IPIs as a shared secret source between legitimate devices attached to the same individual.

The H2K framework operates through a three-stage pipeline that systematically transforms physiological signals into cryptographic keys. The first stage involves heartbeat measurement, where sensors capture either ECG or PPG signals from the user's body. The second stage performs IPI extraction. In this stage, a wavelet transform is first applied for denoising. Then, a peak detection algorithm is used to identify heartbeat peaks. Lastly, an IPI alignment algorithm ensures that both legitimate

devices obtain common IPIs with matching indexes. The third stage implements a key establishment protocol that converts the analog IPI values into digital binary sequences through quantization. Throughout this process, wireless communication modules such as WiFi or Bluetooth facilitate the exchange of timestamps and alignment information without revealing the actual keying material.

Cryptographically, the H2K framework serves as both a key generation and key agreement mechanism for symmetric encryption in multi-device WBAN environments. The system takes ECG signals as input and generates identical cryptographic keys at multiple devices worn by the same individual, effectively transforming physiological signals into a shared secret key distribution channel. These generated keys enable secure communication between WBAN devices through symmetric encryption protocols such as AES. The dynamic and random nature of IPIs ensures that generated keys are unique and time-varying, as no two IPIs are identical, providing inherent defense against replay attacks and key reuse vulnerabilities.

The researchers validated their framework through a comprehensive evaluation using ECG signals from the MIT PhysioBank public database. For ECG evaluation, they simulated two virtual devices running the key generation framework to demonstrate successful key agreement between legitimate parties. To demonstrate practical applicability, the team created a working prototype that successfully encrypted and transmitted an image using AES with heartbeat-generated keys, with the receiving device successfully decrypting the image using its independently generated key.

2) Time-Invariant:

Zhang et al. proposed the ECG-IJS scheme, a biometric-based key agreement protocol for wireless body area networks (BANs) that leverages ECG signals to generate cryptographic keys [18]. By adapting the Improved Juels-Sudan (IJS) fuzzy extractor algorithm to generate a monic polynomial from ECG features, the ECG-IJS scheme addresses critical challenges in BANs without relying on pre-shared keys.

The ECG-IJS scheme operates through a two-phase process of key hiding and key recovery, leveraging the fuzzy vault algorithm. This algorithm works by 'locking' a secret by encoding it into a polynomial. This polynomial is evaluated on a set of values to create genuine points, and then randomly generated "chaff" points are added to hide the genuine ones. This allows only someone with a similar set of values to reconstruct the polynomial and unlock the secret. The IJS fuzzy extractor eliminates the need for chaff points used in traditional fuzzy vault schemes, reducing transmission overhead. The ECG-IJS scheme utilizes this method. In the key hiding phase, the sender first extracts feature F from a 4-second ECG signal segment. The sender then uses these features F as roots to construct a unique monic polynomial of degree s and calculates its coefficients, transmitting only a subset of coefficients to the receiver along with the unencrypted value. In the key recovery phase, the receiver independently extracts features F' from its own ECG measurement from the same human body, constructs a high-degree polynomial using the received coefficients, evaluates this polynomial on all points in F' to generate coordinate pairs, and employs Reed-Solomon decoding to search for a polynomial that satisfies most pairs. The receiver can successfully reconstruct the monic polynomial and recover the secret key K only when F and F' share a sufficient number of common elements, which occurs when both sensors sample ECG signals from the same individual.

The ECG-IJS scheme focuses on secure key agreement. The scheme implements a four-step authentication protocol that begins with key hiding, proceeds through key recovery, validates authenticity through Message Authentication Code (MAC) comparison, and concludes with secure acknowledgment. The sender uses the recovered secret k to encrypt sensitive medical data and compute a MAC for integrity verification. Upon successful key recovery, the receiver decrypts the message and recalculates the MAC; if the calculated MAC matches the received value, authentication succeeds, and the receiver sends an acknowledgement back to the sender.

The researchers validated their approach using actual ECG signals, demonstrating that the feature extraction process provides sufficient discriminative information for reliable key agreement and authentication between legitimate sensors on the same body while preventing unauthorized access from external devices. The parameter t, which determines the number of transmitted coefficients, significantly affects the scheme's performance in terms of False Acceptance Rate (FAR) and False Rejection Rate (FRR), and can be pre-configured in sensor nodes according to specific authentication performance requirements.

González-Manzano et al. proposed Encryption by Heart (EbH), a mechanism that generates time-invariant symmetric encryption keys derived from a user's ECG signal [38]. Unlike previous ECG-based approaches that produce keys that change over time or focus solely on authentication, EbH creates stable encryption keys that remain consistent over extended periods while maintaining uniqueness across users and resistance to prediction attacks.

The EbH system operates through a multi-phase process. Initially, the mechanism constructs a personalized ECG model from reference data collected over 24 hours, establishing a baseline representation of the user's cardiac signature. When encryption is required, the system captures a current ECG scan and extracts relevant feature vectors. These vectors are then compared against the stored model using Walsh-Hadamard Transform coefficients to compute a cryptographic seed. This seed serves as the foundation for generating the symmetric encryption key used to protect the user's data. The critical innovation lies in the decryption process: at any subsequent time, a different ECG scan can be captured, and when its features are compared against the same personal model, the system generates an identical seed. This allows for symmetric decryption using the same

effective key. This approach eliminates the need for physical key storage while ensuring that only the legitimate user, whose current ECG matches the original model, can decrypt the protected information.

Cryptographically, the mechanism enables encryption of sensitive data on demand without manual key management. Its property of time invariance allows for key-based encryption and decryption from a user's real-time ECG.

The researchers evaluated EbH using a comprehensive dataset of 199 subjects monitored over 24 hours. The paper presented that up to 95.97% of users produced unique keys, highlighting that individual users' encrypted data is distinguishable and protected from cross-user attacks. The generated seeds reached lengths of up to 300 bits, providing sufficient keyspace for symmetric encryption algorithms. The system achieved 93.51 bits of min-entropy. The researchers obtained these results by varying parameters such as the observation period and training period lengths, with specific configurations showing trade-offs between key uniqueness, entropy, and stability.

3) Advanced Computational and ML-based Models:

Ye and Huang proposed a chaos-based image encryption algorithm that uses ECG signals to generate initial cryptographic keys for image encryption [39]. The authors specifically address the challenge that digital images differ fundamentally from text due to their bulky data capacity, redundancy, and strong correlation between adjacent pixels, necessitating specialized protection mechanisms beyond traditional text crypto schemes. Their scheme incorporates an autoblocking method that relates the image to the generated keystream to protect against differential attacks.

The encryption algorithm operates through several integrated components. First, ECG signals are processed using the Wolf algorithm to calculate the largest Lyapunov exponent. This Lyapunov exponent is then transformed to generate three initial conditions, two for the generalized Arnold map and one for the logistic map. The algorithm then employs an autoblocking method. Rather than implementing the traditional permutation-plus-diffusion architecture, the scheme focuses exclusively on the diffusion operation. The diffusion process incorporates feedback mechanisms where each cipher block depends on the previous cipher block and includes a control parameter, making the keystream dependent on the plain-image content. The algorithm takes as inputs an ECG signal, control parameters, and a plain image, producing an encrypted cipher image as output.

The scheme is designed to provide robust protection for digital images transmitted over networks. The plaintext-dependent keystream generation specifically addresses known vulnerabilities in previous schemes, such as those identified in algorithms by Vinod Patidar et al., where independence between keystream and plain-image allowed systems to be broken using only one plaintext-ciphertext pair[Patidar]. The authors postulate that by establishing a feedback mechanism from the plaintext to change the keystream for different plain-images, the algorithm can effectively resist known-plaintext, chosen-plaintext, and differential attacks.

The researchers conducted extensive simulations using MATLAB on standard test images, including Lena, Cameraman, and meteorogram images. Security analysis revealed strong differential attack resistance, with UACI reaching 33.456% and NPCR achieving 99.576% after three rounds, approaching the ideal values of 33.3% and 99.5%, respectively. These metrics indicate that a one-bit change in the plain image produces a completely different cipher image. The algorithm also passed comprehensive NIST 800-22 randomness testing across all 15 test items. Key sensitivity tests confirmed that even a one-sampling shift in the ECG signal resulted in complete decryption failure.

Alharbi et al. proposed the Time-Invariant Cryptographic Key (TICK) generation approach to address the intra-individual variance of cardiac signals limitation for ECG signals, which decreases the possibility of obtaining time-invariant keys and increases decryption errors in symmetric cryptography applications [40].

The TICK system operates through a two-phase mechanism. During the enrollment phase, a transformation matrix (W) and quantization thresholds are computed for a set of participants. The transformation matrix W is designed to maximize the ratio of inter-individual to intra-individual variance across participants. When a user requires a cryptographic key, the key generation phase will occur. In this phase, their ECG feature vector undergoes variance improvement by multiplication with the matrix W, producing an improved feature vector. Each feature in the resulting vector is then converted to a binary string using the precomputed quantization thresholds.

The TICK approach is specifically designed for symmetric cryptography applications where key regeneration is essential. The system takes user ECG samples during enrollment and subsequently generates cryptographic keys by determining the variance between different users and between samples of the same user. The quantization method employed in TICK assigns binary strings with equal numbers of 0s and 1s to quantization intervals containing the highest frequency of feature values, thus ensuring enhanced randomness in the final binary sequences. This design consideration is crucial for preventing adversarial guessing attacks while maintaining the time-invariant nature necessary for consistent key regeneration.

Testing was conducted on a multi-session database where samples were collected with an average interval of more than two months between sessions. The researchers report that the approach achieved reliability of up to 96.80% using across-sessions data and up to 98.69% using within-session data with 4-bit quantization. The randomness of generated keys was verified through five NIST statistical tests. Furthermore, computational randomness analysis using min-entropy revealed that the generated keys offered entropy of approximately 1, outperforming comparable approaches [IOMBA].

Kocak et al. proposed a bio-cryptographic system that employs a Siamese Neural Network (SNN) to extract stable 256-bit binary keys from finger-based ECG signals [41], addressing the time-variant challenge through machine learning.

The proposed ECG-based key generation method utilizes a Siamese Neural Network architecture specifically designed to extract binary keys that remain consistent for the same individual while remaining distinct across different individuals. The SNN architecture comprises six 1D convolutional layers, each followed by Max-Pooling operations to downsample the signals while retaining essential features, with Leaky ReLU activation functions applied after each convolution to introduce non-linearity. This deep learning approach was trained using a contrastive loss function that optimizes for high intra-class consistency and inter-class distinctiveness. After the signal passes through the convolutional layers, a fully connected layer yields 256 features, which are thresholded to output into a 256-bit binary key. The system enables reliable key matching through Hamming distance computation without requiring exact key reproduction.

The fundamental premise is that by integrating biometric data with cryptographic techniques, these systems can enhance the security of digital systems. However, it is important to note that the research primarily focused on signal consistency without extensively considering temporal factors, suggesting that expected real-world testing scenarios could produce different results than those presented in the controlled experimental conditions.

The experimental validation of the proposed system was conducted using a public dataset of heartprints obtained from the fingers of 199 users. Entropy measurements approached the theoretical maximum of 1. The system also achieved higher intra-similarity compared to inter-similarity, validating the core objective of the contrastive learning approach. Cross-validation results revealed an average Equal Error Rate (EER) of only 0.02, representing the point where false acceptance and false rejection rates are equal and indicating exceptional discriminative capability. The deep convolutional architecture with multiple layers enabled the network to extract increasingly abstract and invariant features through its hierarchical processing, contributing to the system's ability to generate stable keys despite the inherent variability in biometric signals.

Maldonado Romero et al. also proposed a machine-learning-based cryptographic scheme using ECG [42]. Their system involved a hybrid deep-learning framework for ECG-based cryptographic key generation that combines convolutional neural networks with a transformer-based multi-head attention mechanism to capture broader temporal context.

The proposed framework operates through three distinct phases that transform raw ECG signals into cryptographic keys. In the first phase, the system performs convolutional feature extraction where each standardized ECG segment is processed by a sequence of three 1D convolutional layers using kernel sizes of 15, 10, and 5 and an increasing number of filters. This operation transforms the raw ECG signal into an intermediate feature map. The second phase implements a transformer-based attention mechanism where the output from the convolutional stack is fed into a multi-head attention module. This enables the model to capture dependencies over the entire signal and merge local features with global context. In the third phase, the normalized feature map is condensed using GlobalAveragePooling1D to yield a fixed-length vector, which a dense layer with sigmoid activation maps into a 256-dimensional probability space. A thresholding operation generates the binary key where each bit is set to 1 if the probability exceeds 0.5. The scheme is designed for secure authentication in cybersecurity systems. The generated 256-bit cryptographic keys meet cryptographic standards while preserving critical biometric details, ensuring reproducibility.

The researchers validated their results using the ECG-ID database from PhysioNet. The model was trained using stratified sampling with the Adam optimizer at a learning rate of $1{\times}10^{-}4$, applying binary cross-entropy loss and early stopping with patience of 15 epochs to prevent overfitting. Ground-truth keys were generated using Python's secrets library, serving as the reference benchmark for assessing consistency and reliability. The average intra-person Hamming distance was approximately 46.74 bits, confirming that keys generated from multiple ECG segments for the same subject are highly consistent and ensuring the reliability of the key generation process. The average inter-person Hamming distance, computed using aggregated keys, was approximately 120 bits, demonstrating that keys across different subjects are sufficiently distinct.

IV. Problem Definition

Our capstone sets out to answer three important questions in the space of video-derived cardiac signals and information security. Specifically, can we:

1) Build a real-time rPPG pipeline that accurately extracts a photoplethysmography signal from a consumer-grade video? If so, to what extent does the resolution of standard frame rates limit the precision of the recovered heart rate compared to high-fidelity medical ground truth?
2) Synthesize an ECG signal from rPPG output that is physiologically close enough to a ground-truth ECG to serve as a viable attack input against ECG-based cryptographic schemes? Specifically, to what extent does a GAN-based architecture preserve characteristics such as the Inter-Pulse Interval (IPI) and PQRST morphology necessary to recreate a victim's signature?
3) Demonstrate that ECG-based key generation and authentication protocols are vulnerable to such remotely-derived signals?

To the best of our knowledge and after an extensive literature review, all three problems are currently uncharted territory. We believe that by building on previous works with promising results, there is potential to achieve success in all three problems and provide an accurate assessment of whether the assumed "unobservability" of the ECG remains secure.

V. Limitations

A. Unobservability and Variability in ECG

The most fundamental limitation stems from the fact that ECG signals measure cardiac electrical activity, a physiological phenomenon that cannot be directly observed through non-invasive, remote means. Unlike traditional optical biometrics, ECG's electrical impulses are inherently inaccessible to optical sensors. Our attack pipeline attempts to bridge this observability gap, an approach that relies on statistical associations rather than physical measurements, introducing fundamental uncertainty at each transformation step.

The relationship between facial color changes and ECG is mediated by multiple physiological processes, nearly all of which introduce variability and information loss. Critical ECG features have no direct optical correlation in facial blood flow. While the timing of the R-peak does correspond approximately to the peripheral pulse arrival time, the morphological details that many cryptographic schemes rely upon for entropy extraction are obfuscated. Our CardioGAN attempts to hallucinate these missing features using learned statistical priors from the BIDMC training data, but without direct measurement, the reconstructed signals remain probabilistic estimates.

Cardiac signals exhibit substantial intra-subject and inter-subject variability that significantly complicates both the rPPG extraction and ECG synthesis stages of our attack pipeline. Intra-subject variability arises from numerous physiological and environmental factors. Inter-subject variability presents an even greater challenge for model generalization. Factors such as skin tone, melanin content and other factors can introduce bias that makes our attack more or less effective against certain demographic groups.

B. Latency

The attack pipeline's end-to-end latency of around 40 seconds, while sufficient for proof-of-concept evaluation, may pose practical constraints for real-world deployment against time-sensitive authentication systems. This latency arises from several unavoidable bottlenecks. Firstly, the rPPG extraction stage requires a minimum temporal context window to establish metrics for frequency analysis. Secondly, the CardioGAN stage requires 15-second input segments to generate stable IPI sequences. Finally, both stages include buffering and preprocessing overhead. Many ECG-based authentication protocols are designed to complete within 10-15 second windows to balance security with user convenience. This means our attack pipeline cannot generate synthetic credentials faster than legitimate users can authenticate. This beocmes an issue for time-variant cryptographic schemes, though there are potential solutions, including predictive modeling among others.

C. Dataset Limitations

Dataset limitations represent perhaps the most severe constraint on our work. There only exists one small applicable video to ECG dataset for our use and no rPPG to ECG dataset. As a result, our rPPG evaluation was limited and had to be supported by self-recorded iPhone video clips. The sample size is orders of magnitude below the hundreds of subjects and diverse conditions required for robust evaluation. Similarly, our CardioGAN training relied entirely on the BIDMC dataset, which contains only 53 subjects and is heavily skewed. The BIDMC recordings are from contact sensors with high signal quality, creating a distribution shift when the model encounters noisy rPPG inputs.

D. Inconsistency and Limitations in Existing Work

Our research revealed substantial inconsistencies and gaps in the existing literature on nearly every facet of this project. The rPPG literature reports dramatically different performance metrics across studies. While papers on TS-CAN and PhysNet claim MAE values of 1-4 BPM on specific benchmarks, these results were obtained under tightly controlled laboratory conditions. When we applied these same models to consumer iPhone footage under realistic conditions, performance degraded by an order of magnitude. The CardioGAN literature similarly presents reproducibility challenges. While the original CardioGAN paper demonstrates promising PPG-to-ECG synthesis results, critical implementation details are often omitted or underspecified. Furthermore, the existing CardioGAN work exclusively trains and evaluates on contact PPG from clinical datasets. There is no published research examining generalization to rPPG. Nearly every ECG-based cryptographic scheme posed reproducibility challenges. These schemes did not generally have repositories for their data, code, or evaluation strategies, making it extremely challenging to faithfully recreate the models. Further, there is not comprehensive or standard dataset used. Instead, most cryptographic schemes seemed to cherry-pick the dataset that provided the best results. As a result, achieving similar evaluations with publicly available datasets was a challenge.

VI. Attack Pipeline

A. Overview

The proposed attack pipeline consists of three primary stages that transform remotely captured video footage into synthesized ECG signals capable of compromising ECG-based cryptographic systems. The first stage of the attack begins with video capture

of the target subject, from which rPPG signals are extracted. This process involves facial detection, motion reduction, and pulse extraction. In the second stage, the extracted rPPG signals are input to CardioGAN, a generative adversarial network specifically trained for PPG-to-ECG conversion. In this stage, the generator network transforms the input PPG waveform into synthetic ECG signals against a discriminator network that ensures the generated ECG signals are physiologically plausible. In the third stage, the synthesized ECG output is leveraged to compromise ECG-based cryptographic systems by either replicating biometric authentication templates or predicting cryptographic keys derived from cardiac features, effectively breaking the security assumption that ECG signals cannot be remotely obtained.

B. Threat Model

We assume that a cryptography scheme has been based on user ECG data, and we seek to synthesize data that can fool the system. We make the following assumptions about the adversary:

1) The adversary is assumed to have **black-box access** to the cryptography scheme, meaning their only requirement is to supply synthesized ECG signals as input. In our evaluation, the cryptography schemes were implemented strictly for testing purposes. No internal information about these systems was used when generating the spoofed samples.
2) The commercially available health device market is rapidly growing, spurred by recent initiatives from companies and governments. In this study, we evaluate scenarios where the adversary has access to either PPG or ECG through remote means. We believe the availability of user PPG and ECG data can lead to models that can generate realistic synthetic data because remote PPG can be collected directly from video, which is often publicly available. This synthetic data can reasonably be used to attack health devices, BANs, and integrated medical device ecosystems.

We consider the following primary threats to ECG-based cryptographic systems:

1) **Signal Spoofing:** An adversary synthesizes ECG signals that mimic a target user to bypass authentication or derive cryptographic keys.
2) **Replay Attacks:** Previously captured ECG/PPG signals are reused to gain unauthorized access.
3) **Cross-Modal Inference:** The adversary reconstructs ECG signals from correlated biosignals such as PPG obtained remotely.
4) **Key Recovery:** The adversary derives cryptographic keys from spoofed or reconstructed ECG signals.

C. Threat Actors

We have defined nine potential threat actors that, based on varying rationales, have inherent motivations, capabilities, and abilities. Table I summarizes these threat actors, and further description of each of the actors follows.

Threat Actor	Skill	Physical Access
Intended User	Low	Y
Unintended User	Low	Y
External Opportunist	Medium	N
IT Management	High	N
Device Developer	High	N
Competitor	High	N
Hacker	Med/High	N/Y
Coordinated Group	Very High	N/Y
Nation-State	Very High	N/Y

TABLE I: Risk Assessment of Threat Actors based on capabilities and impact of ECG compromise.

The authorized insider represents a legitimate user with physical access to ECG collection devices that employ cryptographic schemes or authentication systems, such as a patient in a healthcare setting or an employee using biometric security systems [43]. This threat actor possesses authorized physical access to the biometric collection device, along with local or remote system access following successful authentication. Their privileges include access to device documentation and user manuals, the ability to observe operational procedures, handle equipment during routine use, and interact with systems during setup or operation phases. Additionally, they maintain legitimate system credentials that facilitate authorized entry into protected resources. Despite their legitimate status, this actor may be motivated by various factors, including convenience-driven attempts to bypass security safeguards, curiosity regarding their personal biometric data access and control, pursuit of financial or personal benefit through system manipulation, experimental testing of device vulnerabilities, avoidance of organizational monitoring or restrictions, misguided sharing of credentials or devices with unauthorized parties, or frustration-driven non-compliant behavior. This insider threat is significant because their authorized access and system familiarity provide opportunities to exploit vulnerabilities that external attackers cannot readily access, making detection and prevention substantially more challenging than traditional external threat scenarios.

The unintended insider represents an individual with physical proximity to ECG collection devices but lacking authorization to access the device itself or downstream systems it enables [43]. Typical examples include caretakers in medical environments or coworkers in professional settings who operate in close physical proximity to biometric authentication systems. While this actor possesses minimal authorized responsibility for system access following authentication, they maintain physical access to collection devices and often develop an understanding of system capabilities and the resources protected by ECG authentication through routine environmental exposure. Their access profile includes familiarity with device documentation and user manuals, opportunities to observe legitimate usage patterns, and the ability to handle devices during setup or operational phases. However, they lack legitimate system credentials for authentication. As insiders operating within the trust boundary, these actors may be motivated by convenience-driven attempts to bypass security controls, curiosity about biometric data systems, experimental probing of device behavior, deliberate impersonation of authorized users, pursuit of financial or personal gain through unauthorized system access, provision of assistance through improper credential or device sharing, frustration-driven non-compliance with security policies, or perceived entitlement to access based on their physical proximity and operational familiarity. This threat actor profile is concerning because their legitimate presence within the operational environment reduces suspicion while their understanding of system workflows enables sophisticated attack strategies that exploit the gap between physical access and digital authorization.

The external opportunist represents an individual lacking any authorized physical access to ECG collection devices or their associated systems but obtains transient access through opportunistic means [43]. This means can include device loss, theft, physical proximity, or exposure in public or shared environments. Unlike insider threats, this actor holds no legitimate responsibility within the system's operational environment. Their access profile is characterized by opportunistic or transient interaction with devices, including exposed sensors, wireless communication interfaces, or observable device behavior in unsecured settings. Operating entirely outside the trust boundary, external opportunists are primarily motivated by financial gain through system exploitation, impersonation of legitimate users to access protected resources, harvesting of personal health information or elevated privileges for identity theft or fraud, resale or extended unauthorized use of stolen physical devices, or malicious sabotage and disruption of biometric authentication systems. This threat actor may attempt to leverage the inherent vulnerabilities of biometric systems deployed in semi-public or inadequately secured environments. In this way, they can exploit the physical nature of ECG collection devices that must necessarily interface with users in potentially observable contexts, thereby creating attack surfaces that transcend traditional digital security boundaries.

IT management personnel represent a privileged insider threat with administrative responsibility for the infrastructure supporting ECG-based biometric systems [43]. This could include backend servers, authentication services, databases, and access control mechanisms. This threat actor possesses authorized administrative access to systems that store, process, and transmit sensitive biometric data, along with direct management authority over authentication mechanisms, cryptographic key material, user account provisioning, and audit logging infrastructure. Their operational scope encompasses device deployment oversight, system updates, patch management, and incident response coordination. Their access profile includes elevated privileges across backend systems and the capability to modify access controls and authentication policies. Along with this, they may have the authority to alter or disable logging mechanisms, control over cryptographic parameters and key management, and comprehensive knowledge of system architecture and security control implementations. Despite their trusted status, IT management personnel may be motivated by financial gain through data exfiltration or system manipulation, impersonation of legitimate users for unauthorized access, harvesting of personal health information or elevated credentials for sale or exploitation, deliberate sabotage or service disruption, unauthorized surveillance through biometric data access, or intentional weakening and misconfiguration of security controls to facilitate future exploitation [44] [45]. This threat actor profile represents the most critical insider risk as their privileged access enables sophisticated attacks that bypass standard security controls, their administrative authority allows evidence tampering and audit log manipulation, and their intimate system knowledge facilitates attacks that are difficult to detect and attribute within normal operational patterns.

Device developers represent a highly privileged insider threat comprising individuals or teams responsible for the complete lifecycle of ECG collection devices [44]. This threat actor holds inherent control over device architecture, including the design and implementation of biometric sensing hardware, signal processing algorithms, authentication logic, cryptographic implementations, secure key storage mechanisms, and firmware update procedures throughout the device's operational lifespan. Their access profile encompasses the ability to modify access controls, authentication policies, logging mechanisms, and cryptographic parameters at the most fundamental level, coupled with comprehensive knowledge of system architecture and security control implementations. Device developers also possess access to proprietary source code, detailed hardware schematics, low-level debug interfaces, and development environments that expose vulnerabilities before security hardening occurs. This privileged position creates motivation vectors including financial gain through deliberate backdoor implementation or vulnerability introduction for later exploitation, facilitation of user impersonation through authentication bypass mechanisms, systematic harvesting of biometric data through embedded exfiltration capabilities, sabotage through malicious firmware or hardware trojans, unauthorized surveillance enabled by covert data collection features, and intentional weakening of security controls through flawed cryptographic implementations or inadequate key protection [46]. The device developer threat is particularly severe as malicious modifications introduced at the design or manufacturing stage are extraordinarily difficult to detect through conventional security auditing, persist across the entire device population, and fundamentally undermine the

chain of trust upon which all downstream security mechanisms depend.

The competitor represents an external organizational or individual threat actor affiliated with rival entities developing analogous ECG biometric devices or cryptographic systems [43] [44]. These threat actors seek to gain strategic advantage through intelligence gathering on proprietary designs, algorithms, and security vulnerabilities. This threat actor operates entirely outside the authorization boundary with no legitimate responsibility within the target system, though they may concurrently develop comparable technologies through independent research and development efforts. Their capability profile includes substantial technical expertise in biometric signal processing, cryptography, and embedded systems development, legal acquisition of commercial devices through open market purchase, sophisticated reverse engineering capabilities encompassing both hardware and firmware analysis, and proficiency in protocol analysis and authentication behavior characterization. They also may have access to public intelligence sources including technical documentation, patent filings, academic publications, and potentially leaked proprietary materials through industrial espionage or insider collaboration. Competitor motivations center on achieving competitive advantage through intellectual property theft and unauthorized replication of proprietary innovations Through this, they may hope to identify security weaknesses for public notice to damage competitor reputation and market position, accelerate their own internal product development cycles, and achieve financial incentives. This threat actor is particularly significant as their technical sophistication enables comprehensive security analysis, their financial resources support sustained reconnaissance efforts, and their legitimate market presence provides plausible cover for device acquisition and testing activities that would otherwise raise suspicion.

External adversaries represent a variety of unauthorized threat actors ranging from individual hackers to sophisticated nation-state actors, all operating outside the authorization boundary without legitimate system responsibilities and in violation of approved policies and governance frameworks. Individual hackers operate independently without affiliation [44]. They do possess the ability to acquire target devices through purchase, theft, or secondary markets, and they have skills in reverse engineering firmware, hardware probing, protocol analysis, and exploitation of publicly available tools for signal capture, replay attacks, and side-channel analysis. Their motivations include curiosity and exploration of biometric systems, reputation building within hacking communities, financial gain through credential theft or ransomware deployment, user impersonation, ethical security research, or ideological opposition to biometric surveillance technologies. Coordinated hacker groups amplify these capabilities through structured collaboration that pools diverse expertise, resources, and specialized tools across multiple disciplines, enabling attacks beyond individual capabilities [47]. These collectives develop and share custom exploitation tooling, execute multi-stage campaigns with sustained persistence, support criminal markets and gray-market services, and achieve large-scale compromises through coordinated exploitation efforts. At the apex of this threat spectrum, nation-state actors and Advanced Persistent Threat (APT) groups operate under government sponsorship with strategic, political, intelligence, or military objectives as part of broader cyber-operations campaigns [47]. These actors command extensive financial, technical, and human resources enabling access to advanced exploit development capabilities including zero-day vulnerabilities, sophisticated supply chain interdiction, and long-term persistent access mechanisms. Nation-state motivations encompass strategic intelligence gathering and surveillance of targeted individuals or populations, pursuit of military or national security advantages through compromise of defense-related biometric systems, political influence operations and coercion campaigns, long-term geopolitical positioning through infrastructure compromise, and financial benefit through intellectual property theft or economic espionage.

D. Risk Analysis

Threat	**Likelihood**	**Impact**
Signal Spoofing	High	High
Replay Attacks	Medium	High
Cross-Modal Inference	Medium	High
Key Recovery	Medium	Very High

TABLE II: Risk assessment of ECG-based attacks.

Signal spoofing represents the most immediate and practical threat, with high likelihood due to the widespread availability of video data that enables remote extraction of PPG signals. Given the proposed attack pipeline, these signals can be transformed into ECG waveforms capable of bypassing authentication systems. The impact is high, as successful spoofing enables adversaries to impersonate users and potentially derive ECG-based cryptographic keys.

Replay attacks have a moderate likelihood, as they require prior capture of ECG or PPG signals through device compromise, interception, or data leakage. While less flexible than generative attacks, replayed signals can still bypass systems lacking liveness detection, resulting in high impact through unauthorized authentication or access.

Cross-modal inference attacks also present a moderate likelihood. Although they require more sophisticated modeling to reconstruct ECG signals from PPG data, recent advances in generative models make this increasingly feasible. The impact remains high because reconstructed signals can serve as viable inputs to biometric authentication or key derivation systems.

Key recovery represents the most severe outcome, with very high impact due to the potential for complete compromise of cryptographic material derived from ECG signals. While its likelihood is moderate—since it depends on the success of upstream

attacks such as spoofing or inference—it poses a critical risk because it undermines the fundamental security assumption of ECG-based cryptography. Once recovered, keys can be reused indefinitely without requiring continued access to the victim's biometric signals.

Overall, the risk profile is dominated by the combination of high-impact outcomes and increasingly realistic attack vectors enabled by remote sensing and generative modeling. In particular, the accessibility of video-based PPG extraction significantly raises the practical likelihood of end-to-end attacks, making ECG-based cryptographic systems vulnerable even in the absence of direct physical access.

E. Remote PPG Extraction Pipeline

The rPPG extraction pipeline serves as a translator of spatial color variations across video frames to a one-dimensional temporal signal. This process begins with facial landmark detection and tracking to isolate specific Regions of Interest (ROIs), such as the forehead and cheek regions, where blood movement is most detectable on the skin. By spatially averaging pixel intensities within these ROIs—typically prioritizing the green channel for its superior contrast in hemoglobin absorption, we produce a raw trace of the subject's cardiac activity.

To ensure the signal is viable for cross-modal synthesis, we implement motion reduction and signal conditioning to mitigate artifacts from subject movement and ambient light fluctuations. This involves temporal detrending and the application of a 3rd-order Butterworth bandpass filter, restricted to typical physiological heart rate frequencies. The resulting preprocessed waveform provides the necessary physiological basis for the CardioGAN generator, effectively bridging the gap between passive video observation and active signal reconstruction.

F. Remote PPG to ECG Conversion

CardioGAN implements a conditional generative adversarial network (cGAN) architecture specifically designed for cross-modal physiological signal transformation [34]. The system consists of two competing neural networks trained in an adversarial game: a generator network G that learns to synthesize realistic ECG waveforms conditioned on input PPG signals, and a discriminator network D that learns to distinguish between authentic and synthetic pairs of ECG-PPG. The generator takes as input a PPG signal segment and outputs a corresponding ECG signal. Both networks are implemented as deep convolutional architectures. The generator employs an encoder-decoder structure. This structure compresses the input PPG into a low-dimensional representation to capture essential cardiac characteristics. Then, it expands this representation back to full resolution while reconstructing the ECG signal. The discriminator receives both a PPG signal and a candidate ECG signal and outputs a probability score indicating whether the pair is authentic. During training, the generator attempts to maximize the discriminator's misclassification rate, while the discriminator simultaneously adapts to better detect synthetic signals, creating an adversarial arms race that drives both networks toward optimal performance.

Our attack pipeline uses the CardioGAN model to generate ECG from remote PPG data with the goal of maintaining characteristics that are seen in real ECG signals. Through this method, ECG data can be collected from video recordings, allowing for the collection of ECG data to be used to spoof cryptographic schemes.

G. Cryptography Schemes

The resulting ECG signal can then go through postprocessing based on the cryptography scheme that is being spoofed. Different cryptography schemes have varying inputs and resulting outputs. Based on their nature, this ECG signal can then be used immediately, in the case of time-variant schemes, or can be saved for later use, for systems that are time-invariant. While we developed six different cryptography schemes, we only tested our attack pipeline against Heart to Heart and H2K.

H. Real-Time Application

The remote reconstruction stage operates with a relatively small latency in our implementation. This latency is acceptable for most cryptographic attack scenarios, as ECG-based authentication typically requires 10-30 seconds of signal to generate stable keys. The computational footprint is modest enough for real-time execution on an iPhone, requiring approximately 150-250ms per 10-second window on a standard CPU. This is in-line with the recording quality we would expect to be input to the model. An iPhone can record up to 4k at 60 frames per second, which is good enough to get quality rPPG data from.

VII. Methodology

A. Coding Crypto Schemes

While preference was given to models that contained available implementations, six of the eight found models required implementation from their original research papers to test our tool against a variety of studied cryptographic schemes. To ensure faithful and effective recreation of these schemes, we employed a systematic methodology combining generative AI assistance with rigorous validation procedures.

Our implementation process followed a structured six-stage approach designed to maintain fidelity to the original research while producing functional, testable code:

Stage 1: Literature Analysis. Each selected paper underwent a comprehensive review to identify the core algorithmic components. We focused our analysis on extracting mathematical formulations, algorithm pseudocode, and stated implementation constraints from the source material.

Stage 2: Input/Output Specification. We formally defined the input parameters and expected output formats for each scheme based on the paper specifications. This stage established clear interfaces that would enable systematic testing and comparison.

Stage 3: Methodological Decomposition. Complex schemes were decomposed into discrete functional components. Most models were naturally separated into multiple stages, which were mapped to modular code structures for testing and debugging.

Stage 4: Core Algorithm Recreation. Critical algorithmic components were implemented first, including foundational mechanisms such as the Neyman-Pearson lemma, fuzzy vault polynomial, and other scheme-specific cryptographic primitives. These implementations were validated against worked examples from the source papers where available.

Stage 5: Model Integration. Following validation of core algorithms, we assembled complete models according to the methodologies described in their respective papers. Generative AI tools were employed to accelerate initial code generation, but all generated code underwent manual review and refinement to ensure correctness and adherence to the theoretical specifications.

Stage 6: Validation and Iteration. Wrapper functions were developed to execute each implemented scheme on standardized datasets. We compared our results against findings reported in the original papers, including performance metrics, security properties, and correctness measures. When discrepancies were found, we iterated across the entire process until the implementations demonstrated behavior consistent with published results.

Through this systematic approach, six of the eight cryptographic schemes were successfully recreated in Python. All implementations, along with their validation datasets and testing frameworks, are publicly available in the project repository. This methodology ensured that our tool evaluation was conducted against implementations that faithfully represented the studied cryptographic schemes as described in their original research publications.

B. Choosing rPPG tools

Choosing the best rPPG toolset for our use case was a defined process. At first, we believed that using an unsupervised model would best fit our real-time needs, because unsupervised models can generally be more efficient than neural networks in terms of time scale and complexity. We therefore chose CHROM, a color channel processing method that detects color changes in the face of a recorded individual. After testing, we found that the latency involved with CHROM was actually greater than what we expected due to the complexity of running the facial recognition computer vision task on each frame of any recorded video. Since latency was similar with a supervised model compared to an unsupervised model, we explored a suite of rPPG tools instead. To do so, we used rPPG toolbox. While rPPG toolbox does contain a MAE table comparing models and datasets, these models were generally purpose-built for differing video scenarios. As a result, our goal was to determine the effectiveness of CHROM, TS-CAN, and PhysNET on our specific purpose and video data through experimental means. Further analysis is represented in the Evaluation section of this paper.

C. Generating ECG from PPG

In order to generate ECG from the rPPG signals that were achieved from the rPPG toolset, we used CardioGAN [32]. Our CardioGAN model was trained based on the CardioGAN paper. The GAN model was trained for 25 epochs using Compute Unified Device Architecture (CUDA) for the ECG/PPG targets. The BIDMC PPG/ECG dataset was used, and model weights for ECG to PPG and PPG to ECG were concurrently trained. This model utilizes four Adam optimizers, including one for both generators and the GAN loss, one for both generators and the contrastive loss, and one for each discriminator. The Adam optimizers had parameters β_1 =0.5 and β_2 =0.999, the learning rate was set to a 10^{-4} for the first 10 epochs and decayed to 10^{-6} for the ensuing epochs linearly. The GAN loss weights were set to α=3,β=1, γ=100. The resulting data that we obtained after training the model, from the test sampling, are shown in figure 5.

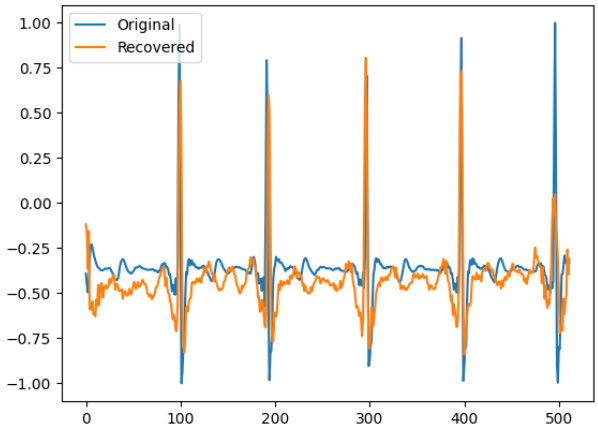


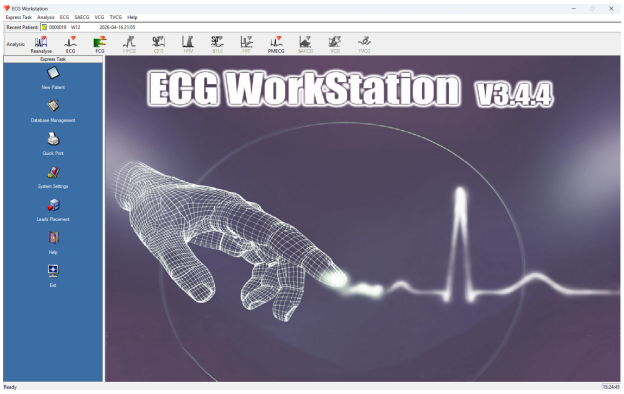
Fig. 5: Example CardioGAN-generated ECG (orange) compared to original ECG (Blue). The generated ECG appears to capture the inter-peak interval morphology of the true ECG well, in addition to reproducing R-peak locations.

D. Getting Ground Truth ECG Data

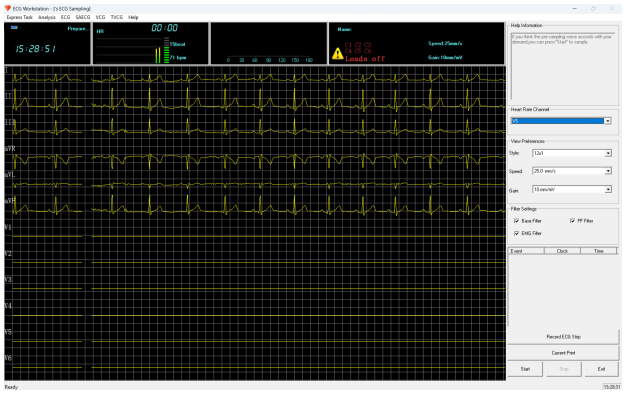
In order to build and evaluate a real-time ECG-spoofing tool, we found it absolutely necessary to obtain high-quality and reliable ground truth ECG data. Our advisor, Dr. Rushanan, provided us with several resources to help us get started. First, he gave us a KardiaMobile (as described in Section 2B). Since the KardiaMobile is only a single-lead ECG device, we would have only used it to cross-check our main recording method. Unfortunately, even though we received it unopened, the device could not connect to our phones via Bluetooth and was rendered useless.

Medical-grade 12-lead ECG devices are difficult to find since they are typically only reserved for clinical use. Thankfully, Dr. Rushanan supplied us with one. The device is called CONTEC8000G ECG Workstation [48]. This device is used by medical and health institutions at all levels to deliver accurate and reliable data for diagnosis. We were also given the accompanying software, ECG Workstation V3.4.4, which is used for real-time display, analysis, and patient database management. Available analyses include ECG, Frequency Cardiogram (FCG), High Frequency ECG, QT internal dispersion analysis, and many others. Only the ECG analysis would be useful for our project.

We first familiarized ourselves with the device, software, and electrode placement. The CONTEC has to connect to a Windows PC via a USB connection and the ECG workstation software will automatically recognize it. 10 electrodes are connected form the device to the body in a set configuration. We placed the electrodes in the standard 12-lead ECG placement which consists of six chest electrodes (V1–V6) and four limb electrodes (RA, LA, RL, LL) designed to capture specific electrical angles of the heart. After exploring the existing software capabilities, we realized that it was insufficient for our use case. While the software produced clean visuals of the 12-lead waveforms and comprehensive statistics such as heart rate, P Interval, QRS Interval, T interval, etc., the output lacked the level of granularity we were looking for. In order to validate the spoofed ECG to the ground truth and pass ECG as input to cryptographic schemes, further analysis at the individual signal level data was required. There was no feature in the software to export the raw heart signals.



(a) Home Screen



(b) Live Reading

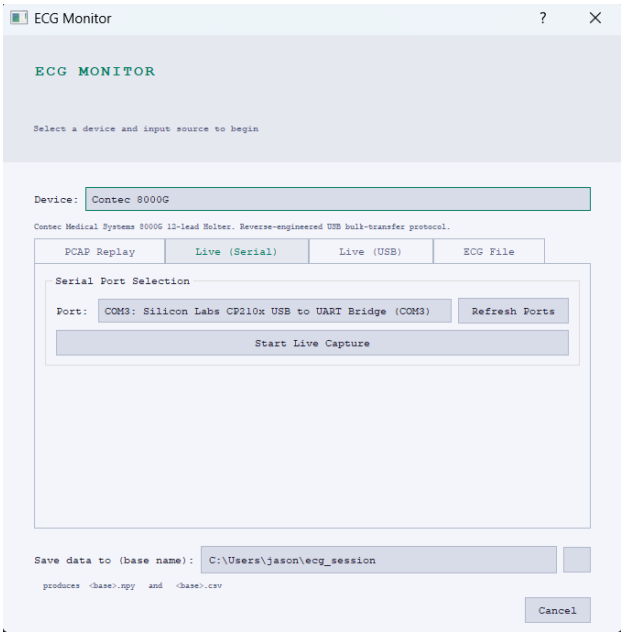
Fig. 6: ECG Workstation V3.4.4 Software

See Appendix A for an example of what the output of an ECG reading of the ECG Workstation V3.4.4 looks like.

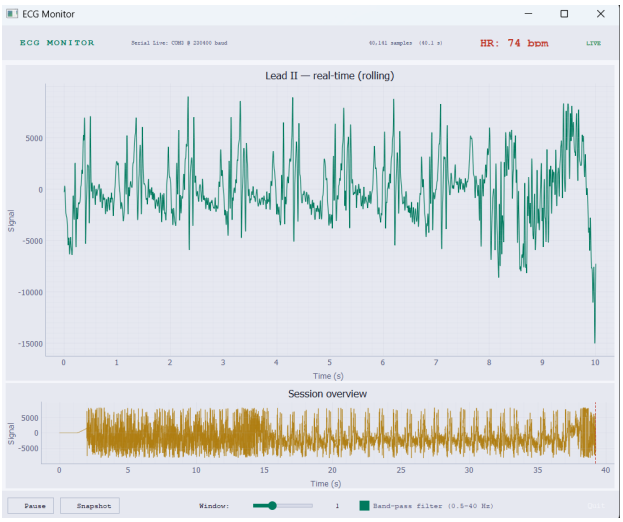
Dr. Rushanan suggested that it may be possible to capture the raw signals as they are being fed into the USB connection. We used Wireshark to sniff the USB traffic from the device to the PC, and by reverse engineering the bytes sent, we were able to uncover exactly how it works under the hood. The communication pipeline starts from the ECG device, to the CP210x Driver, to proprietary DLLs (CP2101.dll), and then finally to the software executable (ECGWorkstation.exe). Instead of opening a standard COM port, the device uses low-level Windows API calls (DeviceIoControl). Before any heart data is sent over, the device sends a complex stream of management commands to the computer to configure leads, filters and safety checks.

1) ECG Monitor App: By reverse engineering the communication between the ECG Workstation device and software, we were able to write our own program that can acquire the heart data, visualize it in real-time, feed it into Python code and store it in formats for easier analysis. We use a Standard Serial Communication to talk to the device at a high speed of 230,400 baud. In our app, we first click ”Start”, which prompts Python to send a 6-step byte sequence to the serial port and wake up the ECG device. Even though the ECG device splits each ECG value across two 7-bit bytes to stay within serial framing rules, the ECG Workstation app puts them back together and applies Two’s Complement calculation to handle negative voltages correctly. We implemented this same logic to our tool. We then filter the raw noise through a Butterworth Bandpass filter (0.5-2.5 Hz) to remove and outlying data. A final touch to make later analysis later is an ”Anchor” timestamp logic where the first sample sent over it matched with the real-world start time and all subsequent samples are spaced exactly 1ms apart, matching the 1000Hz frequency of the ECG workstation.

A lot of the algorithms of the Official ECG Workstation app are still encrypted or proprietary, however we believe that we have built a valuable open-source scientific ECG data collector that is also compatibility with other operating systems. For our future experiments involving comparing synthetic ECG to ground truth ECG data, this app was useful for comparing the raw values.



(a) Home Screen



(b) Live Reading

Fig. 7: ECG Monitor App

E. Server and Real-time Device

We build a small Flask service that the iOS app communicates to over the local network. The phone records a short .mov clip, POSTs it to the server and the server returns a JSON payload with the recovered rPPG waveform, the synthetic ECG, frame rate, and an estimated heart rate. All the computations happen on the host machine, there are no external tools or services called.

Internally the request is processed in four stages, illustrated in Fig. **??**

1) **Decode.** OpenCV reads the clip into an in-memory $T \times H \times W \times 3$ RGB tensor and reports the camera’s native frame rate (typically 30 fps).
2) **Face cropping.** The CHROM path uses a landmark detector (`face_alignment`, SFD backbone) to compute a per-clip convex hull mask over the skin pixels. The two neural paths (TS-CAN and PhysNet) use a Haar cascade on a handful of evenly-spaced frames, take the median bounding box, and resize the cropped face to 72×72.
3) **rPPG extraction.**

- *CHROM*: spatial-mean RGB inside the mask, then the standard chrominance projection.
- *TS-CAN*: 6-channel input (frame-to-frame normalized difference plus the raw $0-1$ scaled patch) fed through the rPPG-Toolbox release weights (`PURE_TSCAN.pth`).
- *PhysNet*: 3-channel normalized-difference input fed through the rPPG-Toolbox release weights (`PURE_PhysNet_DiffNormal`

All three outputs are passed through a 3rd-order Butterworth $[0.7,\ 2.5]$ Hz bandpass and z-scored.

4) **PPG $\rightarrow$ ECG conversion.** The rPPG waveform is polyphase-resampled from the camera rate to 128 Hz, sliced into non-overlapping 4-second (512-sample) windows, min-max scaled to $[-1,\ 1]$ exactly as during CardioGAN training, and pushed through the BIDMC-trained generator `netG_B2A.pth`. The 4-second chunks are concatenated into a single synthetic ECG trace, which is what the iOS app actually plays back.

The server keeps three model objects alive between requests (`face_alignment`, the chosen rPPG model, and the CardioGAN generator), so end-to-end latency on an Apple Silicon laptop is dominated by inference rather than start-up. A 30-second clip processed with TS-CAN takes roughly 30–50 s on CPU, of which face detection accounts for $\sim$2 s.

a) Differences from the toolbox-faithful pipeline of: The deployed server makes several deliberate simplifications relative to the reference `rPPG-Toolbox` implementation that we use for offline evaluation. (i) Face crops are derived per-clip rather than from the toolbox's static $1.5\times$-expanded first-frame box, which makes the system robust to small head shifts but produces a slightly tighter crop. (ii) The TS-CAN raw branch is fed frame$/255$ instead of the toolbox's per-clip z-score normalization. (iii) The post-processing chain omits the toolbox's smoothness-prior detrending and `cumsum` integration step, replacing them with a single 3rd-order $[0.7,\ 2.5]$ Hz Butterworth filter; this is faster and avoids edge artifacts on the short clips the iOS app produces. None of these shortcuts changed the recovered heart rate by more than $\sim$1 BPM on our held-out clips, so the server's output is comparable in accuracy to the reference implementation while running with substantially less code on-device.

F. Evaluation Metrics

To assess the performance of our proposed model and the rPPG extraction pipeline, we employ several complementary evaluation metrics. Together, these provide a robust framework for evaluating both point-wise accuracy and morphological/distributional similarity.

1) Heart Rate and Point-wise Accuracy: We utilize the standard triplet of rPPG metrics to evaluate predicted heart rate ($\hat{f}$) against ground-truth heart rate (f):

- **Mean Absolute Error (MAE):** Represents the average magnitude of the errors in beats per minute (BPM).

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|f_i - \hat{f}_i| \tag{1}$$

- **Root Mean Square Error (RMSE):** Provides a measure of error that penalizes large deviations more heavily than MAE.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(f_i - \hat{f}_i)^2} \tag{2}$$

- **Mean Absolute Percentage Error (MAPE):** Normalizes the error relative to the ground-truth heart rate, expressed as a percentage.

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{|f_i - \hat{f}_i|}{f_i}(\%) \tag{3}$$

Where n is the total number of evaluation windows, f_i is the true heart rate, and $\hat{f}_i$ is the predicted value for the i-th window. We report both MAE and RMSE to distinguish between consistent bias and sporadic outliers; while MAE provides a direct linear interpretation of accuracy, a significantly higher RMSE would indicate the presence of high-variance estimation failures. Together, these metrics allow for a granular assessment of the absolute and relative deviations between our synthesized signals and the hardware-verified ground truth.

2) Morphological and Distributional Similarity: To evaluate the synthesized ECG waveform itself:

The **Pearson Correlation Coefficient (PCC)** measures the linear relationship and morphological similarity between two ECG signals, independent of their absolute amplitudes. For signals $\mathbf{x}$ and $\mathbf{y}$ with means $\bar{x}$ and $\bar{y}$, the PCC is computed as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{4}$$

where $r \in [-1, 1]$, with values closer to 1 indicating stronger positive linear correlation. In ECG analysis, PCC is valuable for assessing whether the generated signals preserve the characteristic shape and temporal patterns of cardiac cycles. Unlike MAE, PCC is invariant to linear transformations of amplitude and offset. A high PCC (typically $r > 0.9$) suggests that the model captures the essential waveform structure and beat-to-beat variability, even if absolute amplitude scaling differs. This

metric is crucial for evaluating whether synthetic ECG signals maintain physiologically plausible cardiac rhythms and wave morphologies.

The **Fréchet Distance (FD)** evaluates the similarity between the statistical distributions of real and generated ECG datasets. Unlike pointwise metrics, FD operates on feature representations extracted from the signals and assesses overall distributional fidelity. Given two Gaussian distributions with means μ_r, μ_g and covariance matrices Σ_r, Σ_g for real and generated data respectively, FD is calculated as:

$$\text{FD} = ||\mu_r - \mu_g||_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right) \tag{5}$$

where $\text{Tr}(\cdot)$ denotes the matrix trace and $||\cdot||_2$ represents the Euclidean norm. Lower FD values indicate that the generated ECG dataset's statistical properties more closely match those of the real data distribution. This metric is particularly important for assessing the diversity and realism of generated ECG signals at the dataset level. This metric captures whether the model merely memorizes the training examples or captures the full variability present in authentic cardiac signals. FD has proven effective in detecting mode collapse and evaluating whether synthetic ECGs span the appropriate range of physiological and pathological conditions represented in the reference dataset [49].

VIII. Evaluation

A. rPPG-Toolbox Evaluation

We evaluated three rPPG methods from the rPPG-Toolbox repository on our own iPhone recordings against ECG ground truth data from the Contec8000G. Our objective here is to examine how accurately can the leading rPPG pipeline recover heart rate from consumer-grade smartphone video. We utilized the exact rPPG-toolbox implementation including preprocessing, model weights, and HR post-processing. This evaluation will shed light on model selection and optimizations for our real-time tool. For each video and ECG pair, we apply the following steps directly taken from the rPPG-toolbox repository:

1) Face crop: Haar cascade detection on the first frame, expanded by LARGE_BOX_COEF = 1.5, applied statically to every subsequent frame and resized to 72x72 with INTER_AREA.
2) Specific model preprocessing:
   - CHROM: raw 72 x 72 RGB frames passed to unsupervised_methods/methods/CHROME_DEHAAN.py.
   - TS-CAN: 6-channel input formed by concatenating the toolbox's DiffNormalized (frame-to-frame derivative scaled by clip-level standard deviation) and Standardized (per-clip z-score) representations.
   - PhysNet: 3-channel DiffNormalized input chunked into 128-frame clips.
3) Each neural model was evaluated under two training datasets (UBFC-rPPG and PURE). Weights used:
   - TS-CAN: UBFC-rPPG_TSCAN.pth, PURE_TSCAN.pth
   - PhysNet: UBFC-rPPG_PhysNet_DiffNormalized.pth, PURE_PhysNet_DiffNormalized.pth
4) HR post-processing: cumsum for derivative-trained models, smoothness-prior detrending with λ = 100, 1st-order Butterworth bandpass over [0.75,2.5] Hz, and FFT periodogram peak-picking inside that band.

Reference HR is derived from a Contec patient-monitor ECG sampled at $\sim$1 kHz. The ECG is read and parsed by out ecg-monitor-app.py described in Section 7B. We synchronize the two clocks by reading the video's creation_time (ffprobe) and the ECG CSV's source_time_unix_s field, and shifting the ECG onto the video timeline. Each video and its ECG are trimmed to a common 5 s warmup followed by a single 25 s evaluation window (lengthened to the longest window that all four clips support).

We compute the reference HR from the R-peak count on the ECG (3rd-order Butterworth [3,45] Hz bandpass, then scipy.signal.find_peaks). This is the medically standard procedure and is what the spoofing detector must ultimately reproduce. We deliberately do not use FFT-envelope HR on the ECG side, because in early experiments both ECG and rPPG FFT estimators locked onto the same low-frequency artifacts, producing falsely optimistic agreement.

We report the performance using the MAE, RMSE, and MAPE metrics as defined in Section VII-G. where $\hat{f}_i$ is the rPPG-predicted HR in window i and f_i is the ECG R-peak HR.

Our test dataset includes four self-recorded iPhone clips, each paired with a Contec ECG session. Subject is stationary, indoors, lit by ambient ceiling light.

TABLE III: Dataset Overview

Clip	Duration	GT HR (ECG R-peaks, BPM)
IMG_2622	41.3 s	70.7
IMG_2623	33.2 s	84.3
IMG_2624	38.4 s	67.4
IMG_2626	41.5 s	86.9

We acknowledge that four windows is below the standard 50–100 used in the original toolbox papers.

TABLE IV: Toolbox-faithful HR estimation error on four iPhone clips, 25 s windows, $[0.75, 2.5]$ Hz band.

Method	MAE (BPM)	RMSE (BPM)	MAPE (%)
CHROM	17.56	18.73	22.07
TS-CAN (UBFC-rPPG)	16.24	17.72	20.31
TS-CAN (PURE)	**10.09**	**11.80**	**12.70**
PhysNet (UBFC-rPPG)	17.56	18.73	22.07
PhysNet (PURE)	18.00	19.30	22.58

1) Findings: TS-CAN trained on PURE is the only method with usable accuracy on our footage. Its MAE of 10.1 BPM and MAPE of 12.7% are within an order of magnitude of the 1–4 BPM MAE these models achieve on the original PURE/UBFC test sets. The other four configurations cluster at near-identical wrong predictions on each clip. On IMG_2624, for instance, CHROM, TS-CAN (UBFC), PhysNet (UBFC), and PhysNet (PURE) all predict 56.2 BPM against a ground truth of 67.4 BPM. Independent algorithms producing identical bias is a strong indicator that the FFT peak-picker is locking onto a shared spectral artifact in the video.

TABLE V: Sensitivity of HR MAE to bandpass low cutoff.

Method	MAE @ $[0.75, 2.5]$ Hz	MAE @ $[1.0, 2.5]$ Hz
CHROM	17.56	25.70
TS-CAN (UBFC)	16.24	18.59
TS-CAN (PURE)	10.09	28.49
PhysNet (UBFC)	17.56	33.76
PhysNet (PURE)	18.00	35.14

The 8.A baseline shows that four of five toolbox-faithful rPPG configurations cannot be relied on to recover heart rate from consumer iPhone footage. Any spoofing attack we evaluate must be measured relative to the appropriate baseline. We use TS-CAN (PURE) as the reference detector throughout the remainder of our evaluations.

B. CardioGAN Evaluation

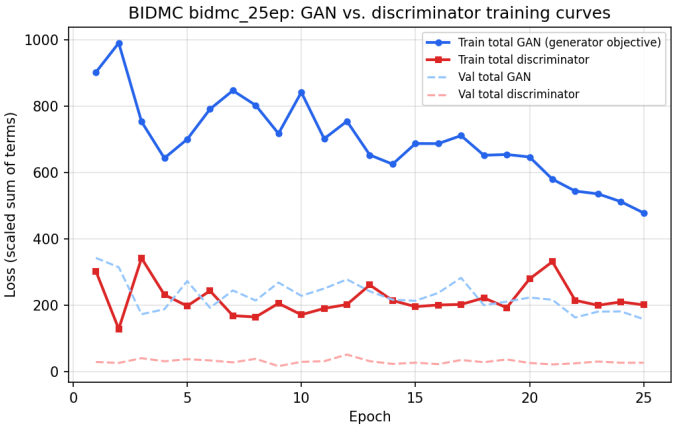


Fig. 8: CardioGAN generator and discriminator loss through training for overall model and evaluation model.

The CardioGAN was trained using the BIDMC dataset. Figure 8 showcases the overall loss of the generator and discriminator of this GAN. From this image, we can see that the GAN was well-trained. In general, we expect that loss for both the generator and discriminator should decrease towards an equilibrium, which need not be the same value, nor should it be 0. The generator loss decreased by about 47% from epoch 1 to 25. Furthermore, the minimum generator loss in the run is at the last epoch. Similarly, the discriminator loss quickly found a broad equilibrium. The evaluation GAN also found equilibria, with the generator loss steadily decreasing and the discriminator loss quickly approaching equilibrium. It is important to note that the loss graph alone suggests that the CardioGAN could be further trained until the generator loss reaches equilibrium. However, as presented later, the CardioGAN could potentially be overfit to the dataset, so further training was not done. Specific metrics shown below showcase that the CardioGAN is effectively trained, but future research could further dissect this dichotomy between under- and over-training. The CardioGAN achieved a heart rate (HR) mean average error (MAE) of 13.629, a Pearson coefficient of 0.0574, and a Fréchet distance of 4.0146 against the test subset of BIDMC, shown in table VI. These values were similar to the train subset of BIDMC, which achieved a heart rate (HR) mean average error (MAE) of 16.08, a Pearson coefficient of 0.0589, and a Fréchet distance of 3.977. The similarity in metrics between test and train showcases that the

model reconstructs and preserves physiology across the generator/discriminator lifecycle.

TABLE VI: CardioGAN Results Across Datasets

Dataset Train/Test	HR MAE	Pearson Coefficient	Fréchet Distance
BIDMC tested on BIDMC	13.629	0.0574	4.015
CAPNO tested on CAPNO	4.94	-0.02	3.60
Capno tested on Own Data	22.915	0.0381	4.67
MCD rPPG/ECG on MCD	23.962	0.2154	4.26

Because our use-case involved real-time data that differed in nature to the PPG in the BIDMC dataset, we also tested the CardioGAN against the CAPNO dataset, which achieved a MAE of 4.94, a Pearson coefficient of -0.02, and a Fréchet distance of 3.60. The comparable results of these two tests are shown in table VI. Based on these results, where the MAE is (higher) and the Pearson is (lower), we can determine that the model is potentially overfit for the BIDMC dataset or that a wide enough variety of data was not used. However, both datasets showcased that the CardioGAN is effective in finding the R-peak locations based on the PPG data, which can be seen in figure 9. For our use case, this fact is most important because many ECG-based cryptography mechanisms utilize IPIs, which directly relate to r-to-r-peak distances. It was also necessary to determine whether the CardioGAN could be trained on rPPG data instead of true PPG data. Thus, we trained the CardioGAN using the MCD dataset, which contains videos and ECG data, after creating rPPG from TS-CAN. This achieved comparable results, a MAE of 23.962, a Pearson coefficient of 0.2154, and a Fréchet distance of 4.26. Interestingly, the CardioGAN was best able to match the signal overal as shown by the pearson coefficient, but the HR was farthest off. This is logical as the rPPG contains its own fairly large HR MAE.

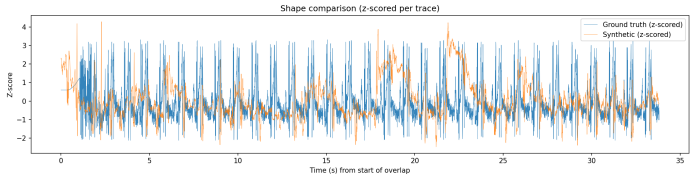


Fig. 9: CardioGAN results from rPPG input data compared to ground-truth.

We also compared the CardioGAN against our own data. Our testing focused on getting ECG data and rPPG data. Our rPPG data was input into the CardioGAN to obtain corresponding ECG data. As discussed previously, the rPPG data has a relatively small MAE for heart rate, so our goal was to determine how effective the CardioGAN ECG generation was against the ground-truth ECG data determined from our sensors. We achieved an MAE of 22.915, a Pearson coefficient of 0.0381, and a Fréchet distance of 4.67, which is shown in table VI. As can be seen from the data, the performance of the model degrades significantly when tested on our own data. Several factors could contribute to this. First, the CardioGAN may need to be trained on a wider variety of data to effectively capture differences in PPG collection. Similarly, it may suggest that rPPG does not effectively capture the signals that the CardioGAN is using to create effective synthetic ECG data. Future research may focus on training a CardioGAN on rPPG data instead of true PPG data to mitigate these potential pitfalls. Even still, the R-peaks seem to be effectively captured, shown in figure 9. As a result, we continued our evaluation on the ECG and cryptography.

C. ECG Evaluation

TABLE VII: ECG Evaluation Metrics and Results

	R-Peaks (#)	IPI Mean (ms)	QRS Mean (ms)	QT Mean (ms)
Ground-Truth	33	1003	81	198
CardioGAN Generated	44	766	80	352

From the CardioGAN evaluation, we know that the generated data had a low Pearson coefficient, signifying that the data was relatively ineffective in capturing the reconstruction of the ECG signal in its entirety. However, the MAE of the heart rate, while not insignificant, showcases some promise that this method is effective in capturing important ECG characteristics. Thus, the next step of our evaluation was to determine the efficacy of our generated ECG data. Several cryptographic models utilize IPI as an ECG characteristic for key generation or security. At the same time, it is important to dissect other factors, including the entire PQRST interval relationships, to determine how these characteristics differ between ground truth and generated. Table VII showcases our results. As can be seen from the data, the pipeline presented is only moderately effective in capturing

the R-peak of an ECG signal from an individual. While it is true that several regions across the ECG signal had matching similarities, there was significant deviation in some regions that heavily skewed the results. For example, there were many more R-peaks detected in the model. This is due most clearly to the artifacts and noise generated by the CardioGAN. The most drastic problem with the generated data seems to be the T curve and peak. Often, this curve was misregistered as the R-peak, and as a result, there was a significant change in QT mean from the ground-truth data and the CardioGAN-generated data. The QRS Mean, on the other hand, was strikingly similar, showing that the relationship of the overall heart rate peak may be effectively captured. Overall, however, it seems like the average error across the entire ECG signal was fairly large.

D. Crypto Evaluation

TABLE VIII: ECG Cryptography Cracking

Dataset Train/Test	H2H Mean Bit Agreement	H2H Key Match (%)	H2K Mean Bit Agreement	H2K Key Match (%)
MCD rPPG/ECG Validation	0.595	9.7	0.634	27.1
CAPNO on Own Data	0.533	0	0.563	10

Our validation testing showcased promising results. We were able to crack 10% and 27.1% of the H2H and H2K algorithms respectively using generated ECG signals from video using the MCD dataset. However, in our real-time testing, these results were not fully replicated. For H2H, the IPI4 bit agreement was around 53.3%, which while greater than average, is not significant to determine efficacy. H2K generates IPIs in a different way than H2H. Rather than using H2H's statistical approach, H2K employs a simpler method to determine the least significant bits of quantized IPIs. While most tests did not authenticate, we did manage to generate matching keys with the H2K scheme using several 15 second windows of ECG data. This signifies that several windows of data did contain matching IPI regions based on this method of R-peak determination. Overall, it is clear that more work needs to be done to move this attack pipeline from the validation stage into real-time application. Based on the previous data, we suspect that the CardioGAN is simply not comprehensive or specific enough to generate the exact necessary data under a variety of data sources. However, this preliminary result does show promise, suggesting that with refinement of PPG extraction and ECG generation methods, these schemes can more consistently be cracked. While several other cryptography schemes were programmed from research papers, our attack pipeline was not tested on them due to time constraints. Thus, future research may continue to conduct testing on other models that use different methods of key generation and agreement.

IX. Discussion

A. Overview

This study investigated the feasibility of compromising ECG-based cryptographic schemes through rPPG for non-contact heart rate extraction from smartphone video and CardioGAN for synthesizing realistic ECG waveforms from PPG signals. While rPPG shows promise as a practical threat vector with the TS-CAN model achieving a heart rate MAE of 10.09 BPM on consumer iPhone footage, the CardioGAN-based attack pathway remains inconclusive due to substantial morphological errors in the generated ECG signals and insufficient training on domain-appropriate data.

The rPPG evaluation revealed that model selection and training dataset composition can be determinants of attack feasibility. Four of five tested configurations demonstrated MAE values clustering around 17-18 BPM, representing errors of approximately 22% MAPE. These models frequently converged on identical erroneous predictions, suggesting systematic locking onto spectral artifacts rather than genuine cardiac signals. TS-CAN proved to be the exception, achieving an MAE of 10.09 BPM. This represents roughly an order of magnitude degradation compared to performance but sufficient for coarse heart rate estimation.

Overall, an error of around 10 BPM represents a credible threat to cryptographic schemes that rely solely on coarse heart rate features or employ overly permissive matching thresholds. The rPPG attack vector's primary advantage lies in its non-contact nature and accessibility. By requiring only a smartphone camera and open-source software, it can become a realistic threat model for adversaries with modest resources.

The CardioGAN evaluation exposed fundamental limitations in translating PPG signals into ECG waveforms with sufficient morphological fidelity for cryptographic attacks. While the model demonstrated stable training dynamics with generator loss decreasing by 47% over 25 epochs and discriminator loss reaching equilibrium, quantitative metrics revealed substantial reconstruction errors. The model achieved an HR MAE of 13.629 BPM and a Pearson correlation coefficient of only 0.0574 on the BIDMC test set, indicating that while the generator produced signals with plausible frequency content, it failed to capture the precise temporal relationships inherent in authentic ECG waveforms.

The issue with generalization became more obvious when evaluating the model on our own iPhone-captured rPPG data. For this set of data, MAE escalated to 22.915 BPM with a Pearson coefficient of 0.0381. This dramatic performance degradation can be attributed to at least three compounding factors. First, the CardioGAN was trained exclusively on high-fidelity contact

PPG from the BIDMC dataset, whereas our attack pipeline provides rPPG signals contaminated by motion artifacts, ambient lighting variations, and compression artifacts from video encoding. The model did not learn the signal transformation from noisy, remote PPG to ECG, creating a fundamental domain mismatch. Second, the relatively small size and limited physiological diversity of the BIDMC training set may have resulted in overfitting; the model appeared to memorize the dataset-specific patterns rather than learning generalizable PPG→ECG mappings. The comparable train and test metrics superficially suggested good generalization within the training distribution, but the results from the out-of-distribution CAPCO data and the rPPG data showcased differently. Third, while the loss curve suggested undertraining, several metrics suggested overfitting to BIDMC characteristics, showcasing a bias-variance trade-off dilemma.

The detailed ECG evaluation revealed that the attack strategy is fundamentally sound. Our validation testing showcased that these systems could be cracked through the spoofing of ECG data from video. However, extrapolation into the full real-time attack pipeline showcased that there were significant morphological distortions in generated signals. While the CardioGAN produced ECG-like waveforms, critical PQRST wave features exhibited large systematic errors. Most notably, the generated signals contained 44 detected R-peaks compared to 33 ground-truth peaks—a 33% false positive rate. This error stems primarily from misclassification of the T-wave as an R-peak, evidenced by the drastically elevated QT interval mean of 352 ms in generated signals versus 198 ms in ground truth. The T-wave misregistration occurs because the CardioGAN seemed to struggle to learn the distinction between the QRS complex and the lower-amplitude T-wave. Without sufficient training examples enforcing these distinctions, the generator produced ambiguous waveforms where multiple peaks exceed detection thresholds.

Conversely, the QRS duration mean was well-preserved, suggesting that the model can successfully learn some local characteristics. This selective preservation indicates that the CardioGAN captures the relative timing of some characteristics while failing to reconstruct others. On the cryptography side, the IPI mean showed substantial deviation, directly reflecting the excess R-peak detections. Many ECG-based schemes, including H2K, derive entropy from precise IPI measurements. As a result, the 237 ms average error in IPI combined with high false-peak rates means that IPI sequences from generated ECG will generally diverged from ground-truth sequences. The increased noise in the generated T-waves corrupted most attempts to authenticate or generate similar keys in these schemes.

The crypto evaluation results reflect the accumulated errors from the entire attack pipeline. While testing of this setup denotes signficant potential, real-time patterns showcased some problems. The H2H scheme, which uses statistical approaches to IPI-based key agreement, showed only 53.3% bit agreement. This near-random performance indicates that the combination of rPPG heart rate error,which was around 10 BPM, and the CardioGAN error combined to obfuscate any signal structure that H2H relies upon for entropy extraction.

The H2K scheme evaluation provided a showcase of some potential vulnerability even in real-time. While it was expected that the H2K algorithm was less secure based on our validation testing, we successfully generated matching keys in several 15-second windows comparing ground-truth ECG signals. This preliminary success demonstrates that when characteristic errors are eliminated, the attack framework can compromise IPI-based schemes. The H2K scheme's simpler quantization method likely provides some inherent robustness to noise, explaining why it was likely more affected by our CardioGAN-based attacks.

This dichotomy suggests that the fundamental attack concept is sound: ECG-based cryptographic schemes can be compromised when an adversary obtains sufficiently accurate cardiac signals. However, the practical realization of this attack via the rPPG→CardioGAN→ECG pipeline remains challenging.

B. Future Work

As previously discussed, the CardioGAN architecture presents multiple opportunities for optimization. First, extending the training duration with more epochs could allow the model to better capture the nuanced patterns and variability inherent in ECG signals. Similarly, some data supported that the model was overtrained for the specific training dataset. Either training the model less or expanding the input dataset with more diverse patient populations could improve the model's generalization capabilities and robustness. Third, incorporating rPPG data as an additional training modality could enhance the quality of generated signals. This third step was deeply considered and researched. However, there are no publicly available rPPG datasets available. The creation of an rPPG to ECG dataset based on video to rPPG datasets was considered and is an avenue of future research.

While CardioGAN demonstrated some promising results, even in the test dataset, metrics like the Pearson Coefficient suggested that ECG-like synthetic signals were not actually created. While this was not essential for our specific purpose, this discrepancy undoubtedly played a large role in the evaluation of our attack pipeline. Transitioning to diffusion-based generative models trained on significantly larger datasets could substantially improve synthesis quality. Diffusion models have shown superior performance in image generation tasks and are increasingly being applied to time-series data. These models excel at capturing complex data distributions and could potentially generate more realistic ECG waveforms. Further, a larger variety of data would more likely generate subtle morphological features important to spoof cryptographic functions. Training such models on large-scale clinical ECG databases could yield synthetic signals that are more indistinguishable from authentic recordings.

The current training objective could be augmented with specialized loss functions that explicitly enforce physiological constraints. Our research involved using a general-purpose CardioGAN. Incorporating domain-specific penalties for accurate

P, Q, R, S, and T peak regeneration would better ensure that generated signals maintain proper cardiac wave morphology. This could include morphological loss terms that penalize deviations in specific intervals, including IPI. Another avenue is to introduce a template-matching loss function that compare generated waveforms against validated cardiac templates. A third potential next step includes frequency-domain constraints to ensure spectral density characteristics are maintained. These physiologically-informed training objectives could improve not only the visual fidelity of synthetic ECGs but also their ability to evade detection.

Due to time constraints, not all developed cryptographic schemes were tested against our attack pipeline. Future research should conduct exhaustive testing across a wider variety of biometric cryptosystems. Investigating cryptographic schemes that use different feature extraction methods, including wavelet transforms, morphological descriptors, deep learning embeddings, or alternative machine learning algorithms could reveal varying levels of vulnerability to synthetic signal attacks. Understanding which cryptographic design choices provide greater resilience would inform the development of more robust biometric authentication systems.

C. Ethical Considerations

This research presents attack pipeline with significant ethical implications that warrant careful consideration. While our stated objective is to identify vulnerabilities in ECG-based cryptographic systems to motivate improved defensive measures, the attack pipeline we describe could theoretically be weaponized by malicious actors. These actors could feasibly compromise medical devices, steal biometric credentials, or circumvent security systems designed to protect sensitive healthcare data and physical infrastructure. The moderate success of the rPPG extraction stage demonstrates that passive, remote cardiovascular monitoring from video is feasible and that an adversary could potentially infer sensitive health information without a victim's knowledge or consent. The application potential towards the cryptographic cracking stage highlights that advancements in model behavior or an increase in publicly available data could lead to further vulnerabilities in ECG-based cryptographic systems.

We conducted this work under the framework of offensive security research. Overall, identifying attack vectors serves the ultimate goal of strengthening defensive systems. The biometric authentication and medical device security communities can only develop robust protections if they understand realistic threat models. Our work demonstrates that the assumption that 'ECG signals are private because they require body contact' (a very commonly seen and discussed position throughout the body of study) is increasingly unrealistic. This is especially true as remote sensing and AI synthesis capabilities advance. We advocate for the implementation of several ethical safeguards moving forward. Firstly, manufacturers of ECG-based authentication and cryptographic systems should be informed of these vulnerabilities. Secondly, regulatory frameworks should be updated to require multi-factor authentication for high-security applications. Thirdly, privacy legislation should explicitly address remote physiological sensing to prevent non-consensual health monitoring via video analysis. Fourth, the research community should establish norms for responsible publication of biometric cryptographic research. We believe that transparent discussion of these vulnerabilities provides the best path toward securing biometric systems before adversaries independently develop more effective attack methods.

X. Conclusion

This research investigated the feasibility of compromising ECG-based cryptographic schemes through a novel attack pipeline combining rPPG and CardioGAN for non-contact ECG synthesis. Our work challenges the long-held assumption that ECG-based biometric authentication is inherently secure due to the presumed inaccessibility of cardiac signals. Further, we demonstrate that meaningful physiological information can be extracted from passive video observation using commodity hardware and open-source software. However, our results reveal a critical gap between theoretical attack concepts and practical exploitation.

The rPPG evaluation demonstrated that state-of-the-art methods exhibit dramatic performance degradation when applied to real-world consumer video compared to laboratory benchmarks. Of five tested configurations, only TS-CAN trained on the PURE dataset achieved usable accuracy on iPhone footage. The CardioGAN evaluation exposed fundamental limitations in translating noisy rPPG-derived signals into ECG waveforms with cryptographically useful fidelity. While the model demonstrated stable training dynamics and achieved moderate performance on in-distribution BIDMC test data, performance degraded on out-of-distribution rPPG inputs.

Our validation testing for our cryptographic evaluation showcased that these systems could be cracked through the spoofing of ECG data from video, with near 10% and 30% cracking rates for H2H and H2K respectively. However, the real-time cryptographic evaluation results reflected the accumulated errors throughout the attack pipeline. The H2H scheme achieved only 53.3% bit agreement, only marginally above random chance and insufficient for authentication. While we successfully generated matching keys against the H2K scheme in isolated 15-second windows using ground-truth ECG signals, the majority of attempts using CardioGAN-generated signals failed to authenticate.

Overall, this research demonstrates that ECG-based cryptographic schemes are fundamentally vulnerable when adversaries obtain sufficiently accurate cardiac signals through any means, but the current state of remote sensing and generative modeling technology introduces errors that may exceed the tolerance thresholds of these schemes.

Our results carry important implications for the design of next-generation biometric authentication systems. System architects should no longer assume that ECG signals are inherently private or unobservable. Defensive measures should take note of this attack avenue and emphasieze features more resistant to remote sensing. Multi-factor authentication combining ECG with complementary biometrics provides defense-in-depth against the possibility that individual modalities may be compromised.

In conclusion, while our attack pipeline in its current form poses limited immediate threat to ECG-based cryptographic schemes, the research demonstrates that the assumption whereupon cardiac electrical signals are inherently private because they require body contact to measure is increasingly unrealistic. By transparently discussing these vulnerabilities while they remain largely theoretical, we aim to catalyze the development of more robust authentication architectures that can withstand the threats of tomorrow's sensor and AI technologies.

APPENDIX A
ECG WORKSTATION SOFTWARE OUTPUT



Fig. 10

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